Chemistry Class XI - Solved Exercises

KPTBB Chemistry Grade XI (Taleem360/Tayarri same book, 419pp) - Complete Book, All 12 Units

Unit 1 — Stoichiometry

Exercise — Choose the Correct Option

Q1. The mass of an atom (element) compared with the mass of one atom of C-12 is called,

a) One mole

b) Gram atomic mass

c) Atomic number

d) Relative atomic mass [Correct]

Q2. Which of the following is not true for a mole?

a) It is a counting unit

b) It is the gram atomic or gram formula mass of a substance

c) It contains 6.023 x 10^23 particles

d) It contains different number of particles for different substances [Correct]

Q3. What is the mass (in grams) of 5 moles of water (H2O)?

a) 90g [Correct]

b) 36g

c) 18g

d) 100g

Q4. The number of molecules in 22g of CO2 is

a) 6.023 x 10^23

b) 3.011 x 10^23 [Correct]

c) 6.023 x 10^21

d) 6.023 x 10^22

Q5. What will be the values of temperature and pressure at standard conditions (STP)?

a) 100 degrees C, 1 atm

b) 298K, 1 atm

c) 273K, 760mm Hg [Correct]

d) 0 degrees C, 760cm Hg

Q6. The molar volume of SO_2 gas at STP is,

a) 64 dm^3

b) 24 dm^3

c) 22.4 dm^3 [Correct]

d) 22.4 cm^3

Q7. The percentage of Ca in $CaCO_3$ is,

a) 12%

b) 10%

c) 48%

d) 40% [Correct]

Q8. Given the equation: $CO_2(g) + C(s) \rightarrow 2CO(g)$. Which of the following equivalences is not correct for the reaction?

a) 1 mol CO_2 = 2 mol CO

b) 1 mol C = 56g CO

c) 44g CO_2 = 28g CO [Correct]

d) 44g CO_2 = 12g C

Q9. A limiting reactant is one

a) Which is present in maximum amount

b) Which produces minimum No. of moles of product [Correct]

c) Which produces maximum No. of moles of product

d) Does not affect the amount of product

Q10. Efficiency of chemical reaction can be checked by calculating

a) Actual yield

b) Theoretical Yield

c) Percentage Yield [Correct]

d) Amount of the reactant unused

Q11. Actual yield will reach the ideal (theoretical) value if the % yield of the reaction is,

a) 10%

b) 50%

c) 90%

d) 100% [Correct]

Q12. The maximum No. of moles are present in

a) 11.2 dm^3 of H2 gas at STP

b) 44.8 dm^3 of N2 gas at STP

c) 67.2 dm^3 of CO2 gas at STP [Correct]

d) 22.4 dm^3 of O2 gas at STP

Exercise — Short Questions

Q1. What is gram atom? Why the concept of gram atom is useful in chemistry?

A gram atom is the amount of an element whose mass in grams equals its atomic mass, and it contains Avogadro's number of atoms. The concept is useful because it lets chemists weigh out a known number of atoms on an ordinary balance and use that fixed number in stoichiometric calculations.

Q2. Explain why balanced chemical equations are used in stoichiometric problems?

A balanced equation shows the exact mole ratio in which reactants combine and products form, because atoms are neither created nor destroyed. These mole ratios are the conversion factors used to calculate

masses, volumes and numbers of particles in stoichiometric problems.

Q3. How will you identify the limiting reagent in a reaction and how it controls the amount of product formed?

Convert each reactant to moles and divide by its coefficient in the balanced equation; the reactant giving the smallest value is the limiting reagent. It is consumed first, so the reaction stops when it runs out, and the amount of product can never exceed what the limiting reagent can produce.

Q4. Why the actual yield is always less than the theoretical yield of a reaction?

The actual yield is usually lower because reactions are often incomplete, some product is lost during separation and purification, and side reactions may consume part of a reactant. The theoretical yield assumes a complete reaction with no losses, which is rarely achieved in practice.

Q5. Can you make the distinction between limiting and excess reagent?

Yes. The limiting reagent is the reactant that is used up first and limits the amount of product formed. The excess reagent is the reactant present in more than the required amount, so part of it is left unreacted at the end of the reaction.

Exercise — Numerical Questions

Q1. The mass of 5 moles of an element X is 60g. Calculate the molar mass of this element. Also, name the element.

Step 1: Molar mass = mass / moles = 60g / 5 mol = 12 g/mol. Step 2: The element with atomic/molar mass 12 is carbon. Answer: 12 g/mol, Carbon (C), as printed in the book.

Q2. Calculate the molecular/formula masses of: (a) C_2H_5OH (b) Al_2O_3 (c) $K_2Cr_2O_7$. (Atomic masses: C = 12, K = 39, Cr = 52, Al = 27, H = 1, O =

16)

Step 1: (a) C2H5OH = (2 x 12) + (6 x 1) + 16 = 46. Step 2: (b) Al2O3 = (2 x 27) + (3 x 16) = 102. Step 3: (c) K2Cr2O7 = (2 x 39) + (2 x 52) + (7 x 16) = 294. Answers: (a) 46, (b) 102, (c) 294.

Q3. Calculate the mass in grams of (a) 7.75 moles of Al2O3, (b) 15 moles of H2SO4, (c) 1.0 x 10^25 molecules of H2O. (Atomic masses: S = 32, Al = 27, H = 1, O = 16) NOTE: the book's printed answers for (b) and (c) do not match the calculation — see the worked steps.

Step 1: (a) Molar mass of Al2O3 = 102 g/mol, so mass = 7.75 x 102 = 790.5g. Step 2: (b) Molar mass of H2SO4 = 98 g/mol, so mass = 15 x 98 = 1470g (the book prints 994g, which does not follow from the given data). Step 3: (c) Moles of H2O = 1.0 x 10^25 / 6.023 x 10^23 = 16.60 mol, so mass = 16.60 x 18 = 298.8g (the book prints 16.60g, which is the number of moles, not the mass). Correct answers: (a) 790.5g, (b) 1470g, (c) 298.8g.

Q4. How many moles are present in each of the following samples? (a) 30g of MgS (b) 75g of Ca (c) 8.85kg of CO2 (d) 40 dm^3 of O2 gas at STP (e) 7.5 x 10^20 molecules of C6H6 (f) 40g of NaCl. (Atomic masses: H = 1, C = 12, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40)

Step 1: (a) MgS = 56 g/mol, so 30 / 56 = 0.5357 mol. Step 2: (b) 75 / 40 = 1.875 mol. Step 3: (c) 8850g / 44 = 201.136 mol. Step 4: (d) 40 / 22.4 = 1.7857 mol. Step 5: (e) 7.5 x 10^20 / 6.023 x 10^23 = 1.2452 x 10^-3 mol. Step 6: (f) NaCl = 58.5 g/mol, so 40 / 58.5 = 0.6837 mol. Answers as printed in the book.

Q5. Calculate the mass of Mg metal required to consume 2560g of CO2 in the reaction: 2Mg(s) + CO2(g) -> 2MgO(s) + C(s). (Atomic masses: C = 12, Mg = 24, O = 16) NOTE: the book prints 'Ans: 2.79g' — the worked steps give 2.79 kg.

Step 1: Moles of CO_2 = 2560 / 44 = 58.18 mol. Step 2: From the equation, 1 mol CO_2 needs 2 mol Mg, so moles of Mg = 2 x 58.18 = 116.36 mol. Step 3: Mass of Mg = 116.36 x 24 = 2792.7g = 2.79 kg. The book's printed '2.79g' is a units error; the correct mass is 2792.7g (2.79 kg).

Exercise — Descriptive Questions

Q1(a). Define and explain mole and Avogadro's number with examples.

A mole is the amount of a substance containing 6.023 x 10^23 particles (atoms, molecules or ions), and this number is Avogadro's number. For example, one mole of carbon (12g) contains 6.023 x 10^23 carbon atoms, and one mole of water (18g) contains 6.023 x 10^23 water molecules.

Q1(b). Given the equation $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g) + Heat$, how can this equation be read in terms of particles, moles and masses?

In particles: 1 molecule of methane reacts with 2 molecules of oxygen to give 1 molecule of carbon dioxide and 2 molecules of water. In moles: 1 mol CH_4 reacts with 2 mol O_2 to give 1 mol CO_2 and 2 mol H_2O. In masses: 16g of CH_4 reacts with 64g of O_2 to give 44g of CO_2 and 36g of H_2O.

Q1(c). Define molar volume of a gas. What will be the volume of 60g of NH_3 at STP? (Atomic masses: N = 14, H = 1)

Molar volume is the volume occupied by one mole of a gas at STP, equal to 22.4 dm^3. Step 1: Moles of NH_3 = 60 / 17 = 3.529 mol. Step 2: Volume = 3.529 x 22.4 = 79.05 dm^3, as printed in the book.

Q2(a). What do you mean by percentage composition of a compound? How the percentage of an element is calculated in a compound?

Percentage composition is the percentage by mass of each element present in a compound. The percentage of an element = (total mass of

that element in one mole of the compound / molar mass of the compound) x 100.

Q2(b). Calculate the percentage composition of each element in: (i) MgSO4 (ii) KMnO4 (iii) NaAl(SO4)2. (Atomic masses: Al = 27, K = 39, Mg = 24, Mn = 55, Na = 23, O = 16, S = 32)

Step 1: (i) MgSO4 = 120: %Mg = 24/120 x 100 = 20%, %S = 32/120 x 100 = 26.67%, %O = 64/120 x 100 = 53.33%. Step 2: (ii) KMnO4 = 158: %K = 24.68%, %Mn = 34.81%, %O = 40.51%. Step 3: (iii) NaAl(SO4)2 = 242: %Na = 9.50%, %Al = 11.16%, %S = 26.45%, %O = 52.89%. These match the book's printed answers (to rounding).

Q3. When steam is passed through red-hot carbon, water gas (a mixture of H2 and CO) is formed: C(s) + H2O(g) -> CO(g) + H2(g). (a) Which is the limiting reagent if 24.5g of carbon is mixed with 1.89 moles of water vapours? (b) Calculate the amount (in grams) of the excess reagent left unreacted. (Atomic masses: C = 12, O = 16, H = 1)

Step 1: Moles of carbon = 24.5 / 12 = 2.04 mol, against 1.89 mol of steam; the equation needs them 1:1, so water is the limiting reagent.

Step 2: Carbon used = 1.89 mol, so carbon left = 2.04 - 1.89 = 0.15 mol.

Step 3: Mass left = 0.15 x 12 = 1.82g (the book prints 1.812g, the difference being rounding).

Q4(a). Calculate the percentage yield if 6.53g of hydrogen gas is produced when 5 mole of zinc is consumed in the reaction: Zn(s) + 2HCl(aq) -> ZnCl2(aq) + H2(g). (Atomic masses: Zn = 65.38, Cl = 35.5, H = 1)

Step 1: From the equation, 5 mol Zn gives 5 mol H2 theoretically. Step 2: Theoretical mass of H2 = 5 x 2 = 10g. Step 3: Percentage yield = (6.53 / 10) x 100 = 65.3%, as printed in the book.

Q4(b). The percentage yield of the following reaction is 85%: 2Al(s) +

3Cl2 -> 2AlCl3(s). How many grams of AlCl3(s) will be actually obtained from 100g of aluminium metal? (Atomic masses: Al = 27, Cl = 35.5)

Step 1: Moles of Al = 100 / 27 = 3.704 mol, which gives 3.704 mol AlCl3. Step 2: Molar mass of AlCl3 = 133.5 g/mol, so theoretical mass = 3.704 x 133.5 = 494.4g. Step 3: Actual mass = 85% of 494.4 = 420.3g (the book prints 420.19g, the difference being rounding).

Q4(c). Given the equation N2(g) + 3H2(g) -> 2NH3(g) at STP, how many moles of NH3 would be formed if 6.3 dm^3 of N2 gas react with an excess of H2? (Atomic masses: N = 14, H = 1)

Step 1: Moles of N2 = 6.3 / 22.4 = 0.281 mol. Step 2: From the equation, 1 mol N2 gives 2 mol NH3, so moles of NH3 = 2 x 0.281 = 0.56 mol, as printed in the book.

Q5. Given the equation 2H2(g) + O2(g) -> 2H2O(g): (a) How many moles of water will be obtained by burning 5.6 moles of O2 in an excess of H2? (b) How many moles of O2 would be needed to react with 58.5g of H2 to form water? (c) How many grams of H2 would be needed to form 120g of H2O? (Atomic masses: O = 16, H = 1)

Step 1: (a) 1 mol O2 gives 2 mol H2O, so 5.6 mol O2 gives 11.2 mol H2O. Step 2: (b) Moles of H2 = 58.5 / 2 = 29.25 mol, and O2 needed = half of that = 14.625 mol. Step 3: (c) Moles of H2O = 120 / 18 = 6.667 mol, so H2 needed = 6.667 mol = 13.33g. Answers as printed in the book.

Project. (1) Take a glass and add 250 cm^3 of water to it. Find out the number of water molecules present in it. (2) Take another glass, add 250 cm^3 of water and then 200g of sugar. Find the number of sugar molecules in the sample and the total number of molecules in the sugar solution.

Step 1: 250 cm^3 of water = 250g = 250 / 18 = 13.89 mol, giving 13.89 x 6.023 x 10^23 = 8.36 x 10^24 water molecules. Step 2: 200g of sugar

(sucrose, molar mass 342 g/mol) = 0.585 mol, giving 3.52 x 10^23 sugar molecules. Step 3: Total molecules in the solution = 8.36 x 10^24 + 0.35 x 10^24 = 8.71 x 10^24 molecules.

Unit 2 — Atomic Structure

Exercise — Choose the Correct Option

Q1. The e/m ratio of cathode rays is,

a) 1.76x10^23 C/kg

b) 1.76x10^11 C/kg [Correct]

c) 1.76x10^-31 C/kg

d) 1.76x10^7 C/kg

Q2. The wave number (vbar) of a radiation with lambda = 2 x 10^8 nm will be,

a) 0.5x10^-8 nm^-1 [Correct]

b) 10 nm^-1

c) 5 nm^-1

d) 100 nm^-1

Q3. For an electron in K-shell, the correct four quantum numbers are represented by,

a) n=1, l=0, m=0, s=+1/2 [Correct]

b) n=2, l=0, m=0, s=+1/2

c) n=1, l=1, m=0, s=-1/2

d) n=2, l=2, m=+2, s=+1/2

Q4. All are electromagnetic in nature except,

a) IR rays

b) X-Rays

c) UV-rays

d) cathode Rays [Correct]

Q5. The energy associated with quantum of radiation is,

a) h nu [Correct]

b) nh / 2 pi

c) 2 h nu

d) (1/2) h nu

Q6. Bohr's theory cannot be applied on,

a) H

b) H+ [Correct]

c) He+1

d) Li+2

Q7. Which of the following sets of quantum numbers are not possible?

a) n=3, l=2, m=0, s=-1/2

b) n=3, l=3, m=-3, s=+1/2 [Correct]

c) n=4, l=3, m=-3, s=+1/2

d) n=3, l=1, m=0, s=+1/2

Q8. The frequency of a green light is 6x10^14 Hz, its wavelength is,

a) 5 nm

b) 500 nm [Correct]

c) 5000 nm

d) 100 nm

Q9. Which of the following sub-shell do not exist?

a) 1p [Correct]

b) 5d

c) 6f

d) 1s

Q10. All quantum numbers are obtained from solution of Schrodinger equation except,

a) Principal quantum number, n

b) Magnetic quantum number, m

c) Spin quantum number, s [Correct]

d) Azimuthal quantum number, l

Q11. The maximum number of electrons in 3s orbital,

a) 1

b) 2 [Correct]

c) 6

d) 10

Q12. The splitting of spectral lines in magnetic field is

a) Stark effect

b) Zeeman effect [Correct]

c) Aufbau principle

d) Pauli exclusion principle

Q13. If the mass of electron is reduced to half, the Rydberg's constant,

a) Becomes half [Correct]

b) Becomes double

c) Remains unchanged

d) Becomes one fourth

Q14. Which element has the electronic configuration of noble-gas notation [Kr] 5s^2, 4d^2?

a) Se

b) Sr

c) Zr [Correct]

d) Mo

Q15. The mathematical form of Moseley's law is

a) sqrt(nu) = b(z-a)

b) sqrt(nu) = a(z-b) [Correct]

c) sqrt(nu) = (a/b) sqrt(z)

d) sqrt(nu) = (b/a) z

Exercise — Short Questions

Q1. Name any four properties of cathode rays.

Cathode rays travel in straight lines away from the cathode and cast sharp shadows of objects placed in their path. They are deflected by electric and magnetic fields, showing they carry a negative charge. They cause fluorescence on a zinc sulphide screen and can blacken a photographic plate. They can penetrate thin metal foils and produce heat when stopped by a metal target.

Q2. How will you differentiate between a continuous and a line spectrum?

A continuous spectrum shows all colours merging into one another without gaps, as given by a hot solid or by sunlight passed through a prism. A line spectrum shows only sharp, separate bright (or dark) lines at fixed wavelengths, as given by excited gases. The line spectrum is characteristic of the element, while the continuous spectrum is not.

Q3. How did Moseley discover that the atomic number (Z) is the fundamental property of an element?

Moseley measured the frequencies of the X-rays given out by different elements and found that the square root of the frequency varies regularly with the atomic number, sqrt(nu) = a(Z - b). Elements fell into a regular order by atomic number, not by atomic mass, and gaps in the sequence predicted undiscovered elements. Atomic number, not atomic mass, therefore fixes an element's place in the periodic table.

Q4. What are the shortcomings of Bohr's atomic model?

Bohr's model works only for one-electron systems such as H, He+ and Li2+ and fails for multi-electron atoms. It cannot explain the fine splitting of spectral lines, the Zeeman and Stark effects, or the shapes of

molecules. It treats the electron as a particle in a fixed orbit, contradicting the wave nature of the electron and Heisenberg's uncertainty principle.

Q5. Can you give the reason that 4s orbital are written first than the 3d orbital.

Orbitals are filled in order of increasing energy, not in order of shell number. The 4s orbital has a lower energy than the 3d orbital because of shielding and penetration effects, so it fills first: the filling order is 4s before 3d. Once filled, the 3d electrons are lower in energy than 4s, which is why 4s electrons are lost first on ionisation.

Exercise — Numerical Questions

Q1. Calculate the distance (Angstrom) between the nucleus and an electron in the 5th orbit of an excited hydrogen atom. Ans.: 13.225 Angstrom

Step 1: The Bohr radius of the nth orbit is r = 0.529 x n^2 Angstrom. Step 2: For n = 5, r = 0.529 x 25 = 13.225 Angstrom. Answer: 13.225 Angstrom, as printed in the book.

Q2. What will be the wave number (vbar) of the spectral line of an electron when it jumps from n2 = 4 to n1 = 2? Ans.: 20.58 x 10^5 m^-1

Step 1: vbar = R(1/n1^2 - 1/n2^2) = 109677 x (1/4 - 1/16) cm^-1. Step 2: vbar = 109677 x (3/16) = 20564 cm^-1. Step 3: Convert to m^-1: 20564 x 100 = 20.56 x 10^5 m^-1 (book prints 20.58 x 10^5 m^-1, the difference being rounding of R). Answer: about 20.58 x 10^5 m^-1.

Q3. What is frequency of a radiation with wave number (vbar) equal to 0.5 x 10^8 m^-1? Ans.: 1.5 x 10^16 Hz

Step 1: Frequency nu = c x vbar, where c = 3 x 10^8 m/s. Step 2: nu = 3 x 10^8 x 0.5 x 10^8 = 1.5 x 10^16 Hz. Answer: 1.5 x 10^16 Hz, as printed in the book.

Q4. Calculate the wavelength of an electron when it moves with the velocity of light. Ans.: 2.424 x 10^-12 m

Step 1: By de Broglie's equation, lambda = h / mv, with h = 6.626 x 10^-34 J s and m = 9.11 x 10^-31 kg. Step 2: lambda = 6.626 x 10^-34 / (9.11 x 10^-31 x 3 x 10^8) = 2.424 x 10^-12 m. Answer: 2.424 x 10^-12 m, as printed in the book.

Q5. What will be the energy (kJ/mol) of an electron residing in n = 3 in hydrogen atom? Ans.: -145.817 kJ/mol

Step 1: Energy of an electron in the nth shell is E = -1312 / n^2 kJ/mol.

Step 2: For n = 3, E = -1312 / 9 = -145.78 kJ/mol (book prints -145.817 kJ/mol, the difference being the value of the constant used). Answer: about -145.8 kJ/mol.

Q6. How much energy is lost when an electron in hydrogen atom jumps from n2 = 3 to n1 = 1? Ans.: 1166.542 kJ/mol

Step 1: Energy lost = 1312 x (1/n1^2 - 1/n2^2) = 1312 x (1 - 1/9) kJ/mol.

Step 2: Energy lost = 1312 x (8/9) = 1166.2 kJ/mol (book prints 1166.542 kJ/mol, the difference being the constant used). Answer: about 1166.5 kJ/mol.

Exercise — Descriptive Questions

Q1(a). What is Planck's quantum theory? What are the postulates of this theory?

Planck's quantum theory states that energy is not emitted or absorbed continuously but in small discrete packets called quanta, each of energy E = h nu. Its postulates are: energy is radiated in whole quanta, never in fractions; the energy of a quantum is proportional to its frequency; and a body can emit or absorb energy only in multiples of h nu. This theory explains black-body radiation and the photoelectric effect, which classical

theory could not.

Q1(b). Prove that E = h c vbar, where E = energy, h = Planck's constant, c = velocity of light, and vbar = wave number.

Step 1: Planck's equation is E = h nu. Step 2: Wave number vbar = 1/lambda, and frequency nu = c/lambda. Step 3: So nu = c x vbar. Step 4: Substituting in E = h nu gives E = h c vbar. Hence proved.

Q1(c). What will be energy of a radiation with lambda = 2 x 10^-8 m?

Ans.: 9.939 x 10^-18 J

Step 1: E = h c / lambda = (6.626 x 10^-34 x 3 x 10^8) / (2 x 10^-8).

Step 2: E = 9.939 x 10^-18 J. Answer: 9.939 x 10^-18 J, as printed in the book.

Q2(a). Write down the main postulates of Bohr's atomic model.

Bohr's postulates are: the electron revolves around the nucleus in fixed circular orbits without radiating energy; its angular momentum is quantised, mvr = n h / 2 pi; energy is emitted or absorbed only when the electron jumps between orbits, and the energy of the photon equals the difference of the two orbit energies, delta E = h nu. The orbits are the stationary states of the atom.

Q2(b). How can Bohr's model of atom be applied to hydrogen atom to calculate the radius of nth shell? (NOTE: the answer printed beside this item in the book, vbar = 1.028 x 10^7 m^-1 and lambda = 9.72 x 10^-8 m, is the 4 to 1 jump value repeated from Q6(c); it is not a radius.)

Step 1: Equate the centripetal force to the electrostatic attraction: mv^2/r = (1/4 pi epsilon0) x (e^2/r^2). Step 2: Apply the quantisation condition mvr = n h / 2 pi and eliminate v. Step 3: This gives r = (epsilon0 h^2 / pi m e^2) x n^2 = 0.529 x n^2 Angstrom. The radius therefore grows as the square of the shell number n.

Q3(a). Derive expression using Bohr's model, for the energy difference

(delta E), frequency (nu) and wave number (vbar) in hydrogen atom.

Step 1: The energy of the nth orbit is E = -2 pi^2 m e^4 / (n^2 h^2), or -13.6/n^2 eV. Step 2: For a jump from n2 to n1, delta E = E2 - E1 = 2 pi^2 m e^4/h^2 x (1/n1^2 - 1/n2^2). Step 3: Since delta E = h nu, the frequency is nu = (2 pi^2 m e^4/h^3) x (1/n1^2 - 1/n2^2). Step 4: Dividing by c gives the wave number vbar = R(1/n1^2 - 1/n2^2), where R is Rydberg's constant.

Q3(b). How does Bohr's model explain the hydrogen spectrum?

When hydrogen is excited, its electron jumps to a higher orbit; on falling back it emits a photon whose energy equals the difference of the two orbits. Because the orbit energies are fixed, only certain frequencies are emitted, giving sharp lines instead of a continuous spectrum. Jumps ending at n = 1 give the Lyman series, at n = 2 the Balmer series, at n = 3 the Paschen series, and so on.

Q4(a). What are X-rays? How are these rays produced?

X-rays are high-energy electromagnetic radiations of very short wavelength (about 0.01 to 10 nm), produced when fast-moving electrons strike a metal target. In an X-ray tube, electrons from a heated cathode are accelerated by a high voltage and hit a tungsten anode; their sudden stopping produces continuous X-rays, and the knocking out of inner-shell electrons followed by electron falls produces characteristic X-rays.

Q4(b). Enlist some characteristics of X-rays.

X-rays travel in straight lines at the speed of light and are not deflected by electric or magnetic fields. They penetrate matter, the penetration increasing with their energy, and they are stopped more by dense materials such as bone and lead. They blacken photographic plates, cause fluorescence, and ionise gases, which is why they are used in medical imaging and in crystallography.

Q5(a). Define quantum numbers. What information is given by the spin quantum number? What are the possible values for this quantum number?

Quantum numbers are the four numbers that completely describe an electron in an atom: principal (n), azimuthal (l), magnetic (m) and spin (s). The spin quantum number tells whether the electron spins clockwise or anticlockwise about its own axis, giving it a magnetic moment. Its only possible values are +1/2 and -1/2, so an orbital can hold at most two electrons of opposite spin.

Q5(b). What information is given by the magnetic quantum number?

The magnetic quantum number (m) tells the orientation of an orbital in space and the number of orbitals in a sub-shell. For a given l it takes the values from -l through 0 to +l, that is 2l + 1 values. It also explains the splitting of spectral lines in a magnetic field, the Zeeman effect.

Q6(a). What is an orbital? How does it differ from an orbit?

An orbital is the three-dimensional region around the nucleus where the probability of finding an electron is maximum, about 90 to 95 percent. An orbit is Bohr's fixed circular path of definite radius, while an orbital has no sharp boundary and follows from the wave-mechanical model. An orbit holds electrons in a definite path; an orbital holds at most two electrons and has a definite shape and energy.

Q6(b). Discuss the shapes of s, p and d orbitals.

The s orbital is spherical and non-directional, its size growing with n. The p orbitals are dumb-bell shaped with two lobes on opposite sides of the nucleus, one each along the x, y and z axes (px, py, pz). The d orbitals have four lobes in most cases (dxy, dyz, dzx, dx2-y2), while dz2 has two lobes along the z-axis with a ring around the centre.

Q6(c). Calculate the wave number and wavelength of a photon when

electron jumps from n2 = 4 to n1 = 1. Ans.: vbar = 1.028 x 10^7 m^-1, lambda = 9.72 x 10^-8 m

Step 1: vbar = R(1/1 - 1/16) = 109677 x (15/16) = 102822 cm^-1. Step 2: Convert: 102822 x 100 = 1.028 x 10^7 m^-1. Step 3: lambda = 1/vbar = 1 / (1.028 x 10^7) = 9.72 x 10^-8 m. Answers as printed in the book.

Q6(d). Identify the series of spectral lines to which this photon belongs?

The electron falls to n1 = 1, the innermost shell, so the line belongs to the Lyman series, which lies in the ultraviolet region.

Project

Project 1. Take three different salts and identify unknown metal ion in them by using flame test.

Step 1: Clean a platinum wire in concentrated hydrochloric acid and heat it until it gives no colour. Step 2: Dip the wire in the salt paste and hold it in the non-luminous flame. Step 3: Note the flame colour and identify the metal: sodium gives golden yellow, potassium lilac, calcium brick red, barium apple green and copper bluish green. Step 4: Repeat for each of the three salts and record the colours in a table.

Project 2. Construct a simple gas discharge tube from low cost substances, showing the electrodes, vacuum pump, power supply etc.

Step 1: Take a glass tube fitted with two metal electrodes and a side tube connected to a vacuum pump. Step 2: Fill the tube with a gas at low pressure, about 0.01 mm Hg, using the pump. Step 3: Connect the electrodes to a high-voltage power supply and switch it on. Step 4: Observe the coloured glow of the discharge and label the cathode, anode, vacuum connection and power supply in your diagram.

Unit 3 — Theories of Covalent Bonding and Shapes of

Self-Assessment

Q1. What are the main features of molecular orbital theory?

The main features of molecular orbital theory are: atomic orbitals of similar energy combine to form molecular orbitals spread over the whole molecule; the number of molecular orbitals formed equals the number of atomic orbitals combined; molecular orbitals are of bonding, antibonding and non-bonding types; electrons fill them in order of increasing energy following Aufbau, Pauli and Hund rules; and bond order = (bonding electrons - antibonding electrons) / 2 decides stability and magnetism.

Q2. Explain the molecules of Helium and Oxygen on the basis of MOT.

Helium has 4 electrons in total, so He2 would place 2 electrons in the bonding sigma(1s) orbital and 2 in the antibonding sigma*(1s) orbital, giving bond order zero, so He2 does not exist. Oxygen has 16 electrons; its molecular orbital filling puts 2 unpaired electrons in the pi* antibonding orbitals, giving bond order 2 and explaining why O2 is paramagnetic and has a double bond.

Q3. What is the significance of MOT?

MOT is significant because it explains what valence bond theory cannot: the paramagnetism of oxygen, the non-existence of He2, bond orders that are fractions, and the electronic spectra of molecules. It also predicts bond strength and magnetic behaviour from a single electron-filling scheme.

Q4. Define bond length and ionic character of a molecule.

Bond length is the average distance between the nuclei of two atoms joined by a chemical bond, usually measured in picometres or Angstrom. Ionic character is the partial ionic nature of a covalent bond, arising when the shared electrons are pulled more towards the more electronegative

atom, giving it a partial negative charge and the other atom a partial positive charge.

Q5. Write down the different application of dipole moment.

Dipole moment is used to find molecular shape (a zero value points to a symmetrical shape such as linear $BeCl_2$ or planar BF_3), to measure the ionic character of a bond, to distinguish cis and trans isomers, and to predict solubility and boiling points, since polar molecules attract one another more strongly.

Q6. Explain the effect of bonding on physical and chemical properties.

Bonding changes physical properties such as melting point, boiling point and hardness, because stronger bonds need more energy to break. It changes chemical properties such as reactivity and the type of reaction a substance undergoes, since ionic, covalent and metallic bonding give very different behaviour towards solvents, heat and electricity.

Q7. Explain the change in bond lengths of hetero-nuclear molecules due to difference in electronegativity values of bonded atoms.

For a heteronuclear molecule the observed bond length is shorter than the sum of the covalent radii of the two atoms. The difference grows as the electronegativity difference of the two atoms grows, because the bond gains partial ionic character, and the extra electrostatic attraction pulls the nuclei closer together.

Q8. Explain the difference between molecular, network and metallic solid.

Molecular solids are made of separate molecules held by weak forces, so they are soft with low melting points, like ice and iodine. Network solids have atoms joined by covalent bonds into a continuous three-dimensional structure, so they are very hard with very high melting points, like diamond. Metallic solids have positive ions in a sea of free electrons, so they conduct heat and electricity and are malleable and ductile.

Exercise — Choose the Correct Option

Q1. The head on overlap of p-orbitals of two atoms give rise to bond called,

a) Sigma bond [Correct]

b) Pi (pi) bond

c) Ionic bond

d) Metallic bond

Q2. Which one of the following molecules has a pyramidal structure?

a) CH4

b) NH3 [Correct]

c) H2O

d) C2H4

Q3. Specie in which the central atom uses sp^2 hybrid orbitals in its bonding is,

a) PH3

b) NH3

c) CH3+ [Correct]

d) SbH3

Q4. On the basis of VSEPR theory, a molecule with three bond pair and one lone pair of electrons will have a structure

a) Linear

b) Trigonal planar

c) Tetrahedral

d) Trigonal pyramidal [Correct]

Q5. Which of the following is an example of a network solid?

a) Sugar, C12H22O11(s)

b) Graphite, C(s) [Correct]

c) Carbon dioxide, CO2(s)

d) Magnesium fluoride, MgF2(s)

Q6. All are true for pi-bond except

a) pi-bond is formed from sp hybrid orbitals [Correct]

b) pi-bond is formed by the parallel overlap of half filled p-orbitals

c) pi-bond is formed when a sigma bond is already present

d) pi-bond is weaker than sigma bond

Q7. Which one of the following molecules has zero dipole moment

a) NH3

b) NF3

c) BF3 [Correct]

d) H2O

Q8. The bond angle in H2O is

a) 90 degrees

b) 109.5 degrees

c) 180 degrees

d) 104.5 degrees [Correct]

Q9. Which one of the following compounds has the highest ionic character

a) HF [Correct]

b) HCl

c) HBr

d) HI

Q10. Liquid oxygen is

a) Diamagnetic

b) Paramagnetic [Correct]

c) Antimagnetic

d) Ferromagnetic

Q11. In Carbon dioxide (CO2) molecule, the number of sigma covalent bond(s) is/are,

a) 1

b) 2 [Correct]

c) 3

d) 4

Q12. H2S has a net dipole moment while BeF2 has zero dipole moment, because

a) H2S molecule is linear while BeF2 is angular

b) H2S molecule is angular, while BeF2 molecule is linear [Correct]

c) Fluorine has more electronegativity than S

d) Be is more electronegative than S

Q13. Which one has the least bond angle

a) NH3

b) H2O [Correct]

c) CH4

d) BF3

Q14. The bond order in O2 molecule is

a) Zero

b) 1

c) 2 [Correct]

d) 3

Q15. The unhybridized 'p' orbital in sp^2 hybridization is

a) parallel to sp^2

b) in the same plane

c) perpendicular to sp^2 orbitals [Correct]

d) out of plane

Exercise — Short Questions

Q1. What evidence can you find that the CO2 molecule has linear but H2O

molecule has angular structure?

The evidence comes from dipole moments. Carbon dioxide has a zero dipole moment, so its two equal C=O bond dipoles cancel exactly, which is possible only if the molecule is linear. Water has a net dipole moment, so its two O-H bond dipoles do not cancel, which shows the molecule is bent or angular.

Q2. Give the reason that why O2 molecule is paramagnetic.

On the molecular orbital diagram of O2, the last two electrons go singly into the two pi* antibonding orbitals, following Hund's rule. Two unpaired electrons make the molecule attracted to a magnetic field, which is why liquid oxygen is paramagnetic.

Q3. What ideas justify that the bond angle in NH3 is 107.5 degrees but in H2O, it is 104.5 degrees?

Both shapes follow from lone-pair repulsion in VSEPR theory. Ammonia has three bond pairs and one lone pair, and the lone pair pushes the N-H bonds together from 109.5 degrees to 107.5 degrees. Water has two bond pairs and two lone pairs, and the two lone pairs repel still more strongly, closing the angle further to 104.5 degrees.

Q4. Differentiate between atomic and molecular orbitals.

An atomic orbital belongs to a single atom and is centred on one nucleus, while a molecular orbital belongs to the whole molecule and is spread over two or more nuclei. Atomic orbitals combine to form molecular orbitals, which are either lower in energy (bonding) or higher in energy (antibonding) than the parent atomic orbitals.

Q5. What data was used to make the conclusion that the calculated bond lengths of the heteronuclear molecules are always higher than experimental values?

The data used were the experimental bond lengths measured for

heteronuclear molecules compared with the sums of the covalent radii of the two atoms. The observed lengths were always shorter than the calculated sums, and the shortening grew with the electronegativity difference, showing that the bonds had partial ionic character pulling the nuclei closer.

Q6. How would you prove that the sigma (sigma)-bond is stronger than pi (pi)-bond?

A sigma bond is formed by head-on overlap, so the electron density lies directly between the two nuclei on the bond axis and the overlap is large. A pi bond is formed by sideways overlap, so the density lies above and below the axis and the overlap is smaller. The larger overlap makes the sigma bond shorter and stronger, and a pi bond can only exist alongside a sigma bond, never alone.

Q7. What information would you use to support the view that the molecule of the He2 does not exist?

The molecular orbital filling of He2 would put two electrons in the bonding sigma(1s) orbital and two in the antibonding sigma*(1s) orbital. Bond order = (2 - 2) / 2 = 0, so no net bond exists and the molecule cannot be stable. Helium therefore exists as single atoms, never as He2 molecules.

Q8. Differentiate between bonding and anti-bonding orbitals.

A bonding molecular orbital is lower in energy than the parent atomic orbitals and has increased electron density between the nuclei, holding them together. An antibonding molecular orbital is higher in energy, has a node between the nuclei, and pushes them apart. Electrons in bonding orbitals stabilise a molecule; electrons in antibonding orbitals destabilise it.

Q9. Give reason that why the structure of BF3 is trigonal planar and NH3 is trigonal pyramidal.

Boron in BF3 has three bond pairs and no lone pair, so the three pairs spread as far apart as possible in one plane, giving a trigonal planar shape with 120 degree angles. Nitrogen in NH3 has three bond pairs and one lone pair, and the lone pair pushes the three N-H bonds downwards into a trigonal pyramidal shape with a 107.5 degree angle.

Q10. Why the bond energies of polar molecules are higher than the non-polar molecules?

A polar bond has partial ionic character, so besides the shared-electron attraction there is an extra electrostatic attraction between the partial charges. Breaking such a bond must overcome both the covalent sharing and this ionic attraction, so polar molecules have higher bond energies than comparable non-polar molecules.

Exercise — Descriptive Questions

Q1(a). Draw a Lewis structure for AlCl3, PH3 and H2O. Use VSEPR theory to predict the molecular shape of these molecules.

Lewis structures: AlCl3 has Al with three single bonds to Cl and no lone pair on Al; PH3 has P with three P-H bonds and one lone pair; H2O has O with two O-H bonds and two lone pairs. By VSEPR, AlCl3 (3 bond pairs, 0 lone pairs, AX3) is trigonal planar; PH3 (3 bond pairs, 1 lone pair, AX3E) is trigonal pyramidal; H2O (2 bond pairs, 2 lone pairs, AX2E2) is bent or angular.

Q1(b). What VSEPR notations correspond to molecules that have a bent shape?

Bent shapes come from the VSEPR notations AX2E (two bond pairs and one lone pair, as in SO2) and AX2E2 (two bond pairs and two lone pairs, as in H2O). In both cases the lone pairs push the two bond pairs together, so the molecule is bent rather than linear.

Q1(c). Discuss the different types of overlapping that lead to sigma bond formation.

Sigma bonds form by head-on overlap in three ways: s-s overlap, as in H2; s-p overlap, as in HCl; and p-p head-on overlap, as in F2. In every case the overlapping orbitals meet along the line joining the two nuclei, so the electron density is concentrated on the bond axis.

Q2(a). Explain how sigma and pi bonds are similar and how they are different.

Sigma and pi bonds are similar in that both are covalent bonds formed by the overlap of atomic orbitals and both hold atoms together. They differ in that a sigma bond forms by head-on overlap with electron density on the bond axis and allows free rotation, while a pi bond forms by sideways overlap with density above and below the axis, is weaker, and prevents free rotation about the bond.

Q2(b). Define and explain resonance with suitable example.

Resonance is the description of a molecule by two or more Lewis structures that differ only in the positions of electrons, not of atoms, when no single structure explains its properties. For example, ozone (O3) is drawn with the double bond on either side, and the real molecule is a hybrid with two equal O-O bonds, each intermediate between a single and a double bond.

Q2(c). Discuss the (i) Bond energies (ii) Bond length and (iii) Ionic character with suitable examples.

Bond energy is the energy needed to break one mole of a bond, for example about 436 kJ/mol for H-H. Bond length is the average distance between two bonded nuclei, for example 74 pm for H-H, and it decreases as bond order increases: a triple bond is shorter and stronger than a double bond, which is shorter and stronger than a single bond. Ionic

character grows with the electronegativity difference, as in HF, where the bond is strongly polar, unlike the pure covalent bond in H2.

Q3(a). Explain the HCl, H2O and N2 molecules on the basis of valence bond theory.

By valence bond theory, HCl forms by s-p overlap of the H 1s orbital with a Cl 3p orbital, giving one sigma bond. H2O forms by overlap of the two H 1s orbitals with two half-filled O 2p orbitals, giving two sigma bonds and a bent shape. N2 forms by one sigma bond from head-on p-p overlap plus two pi bonds from sideways p-p overlap, giving a triple bond.

Q3(b). Define and explain hybridization with reference to sp mode of hybridization.

Hybridization is the intermixing of atomic orbitals of similar energy to form an equal number of new hybrid orbitals of the same shape and energy. In sp hybridization, one s and one p orbital mix to give two sp hybrid orbitals pointing in opposite directions, 180 degrees apart, as in BeCl2. The two unhybridized p orbitals remain perpendicular and can form pi bonds, as in ethyne.

Q3(c). Describe how knowledge of molecular polarity can be used to explain some physical and chemical properties of molecules?

Molecular polarity explains many properties: polar molecules have higher boiling points than non-polar molecules of similar mass because of dipole-dipole attraction; polar substances dissolve in polar solvents while non-polar substances dissolve in non-polar solvents; and polarity decides the geometry evidence, since a zero dipole moment shows a symmetrical shape. The dipole moment is therefore a practical tool for predicting behaviour.

Q4(a). Write down the main points of molecular orbital theory.

The main points of molecular orbital theory are: atomic orbitals combine

to form molecular orbitals equal in number to the atomic orbitals used; bonding orbitals are lower in energy and antibonding orbitals higher than the parent orbitals; electrons fill molecular orbitals in increasing energy order following Aufbau, Pauli and Hund rules; and bond order and magnetism follow from the filling. The theory explains stability, bond strength and magnetic properties together.

Q4(b). Draw the bonding in the molecule of oxygen with the help of molecular orbital theory.

The oxygen molecular orbital order is sigma(1s), sigma*(1s), sigma(2s), sigma*(2s), sigma(2pz), pi(2px) = pi(2py), pi*(2px) = pi*(2py), sigma*(2pz). Filling 16 electrons gives the configuration up to pi*(2px)1 pi*(2py)1, with the last two electrons unpaired in the two pi* orbitals. The diagram is drawn with bonding levels below and antibonding levels above their parent atomic levels.

Q4(c). Explain with the help of MO theory, the paramagnetic character of oxygen.

In the molecular orbital filling of O2, the two highest-energy electrons occupy the pi*(2px) and pi*(2py) antibonding orbitals singly, not paired, because Hund's rule keeps them in separate orbitals. These two unpaired electrons give oxygen its paramagnetic character, which valence bond theory, with all electrons paired in a double bond, cannot explain.

Q5(a). How the geometry of the molecule can be determined by using the dipole moment.

The geometry of a molecule can be read from its dipole moment: a zero dipole moment means the bond dipoles cancel, which happens only in a symmetrical shape, such as linear CO2 or BeCl2, planar BF3 or tetrahedral CCl4. A non-zero dipole moment means the dipoles do not cancel, so the shape is unsymmetrical, such as the bent H2O or pyramidal

NH3.

Q5(b). How could you measure the dipole moment of a molecule?

Dipole moment is measured from the dielectric constant of the substance. A polar gas or liquid placed between charged plates changes the capacitance, and from the measured dielectric constant the dipole moment is calculated by the Debye equation, mu = charge x distance. For gases the measurement is made at several temperatures and the orientation part separated from the induced part.

Q5(c). Write down the difference among molecular, network and metallic solids.

Molecular solids consist of separate molecules held by weak intermolecular forces, so they are soft with low melting points, like ice and sugar. Network solids have atoms joined by strong covalent bonds throughout the crystal, so they are very hard with very high melting points, like diamond and quartz. Metallic solids have metal ions in a sea of delocalised electrons, so they conduct electricity and heat and can be hammered into sheets and drawn into wires.

Project

Project. Use ball and stick models to represent the molecules of BF3 and NH3 in front of your class.

Step 1: Take balls of different colours for B, F, N and H, and sticks for bonds. Step 2: For BF3, join three F balls to the central B ball in one plane at 120 degrees to each other, showing the trigonal planar shape. Step 3: For NH3, join three H balls to the central N ball pointing downwards, leaving room above for the lone pair, showing the trigonal pyramidal shape. Step 4: Present both models to the class and point out how the lone pair changes the shape.

Unit 4 — States of Matter 1: Gases

Exercise — Choose the Correct Option

Q1. If absolute temperature of a gas is doubled and the pressure is reduced to, one-half the volume of gas will,

a) remain unchanged

b) double

c) reduce to half

d) increase four times [Correct]

Q2. One dm^3 of Hydrogen at STP weighs approximately

a) 0.0789g

b) 0.0799g

c) 0.0987g

d) 0.0899g [Correct]

Q3. In a factory producing liquid air, one of the pipes carrying dry air at -80 degrees C is blocked with a white solid. This white solid is,

a) Argon

b) Ice

c) Nitrogen

d) Carbon dioxide [Correct]

Q4. The spreading of perfume or scent in air is due to

a) Diffusion [Correct]

b) Effusion

c) Attraction with air

d) Low density

Q5. A gas has certain volume at 10 degrees C. How much temperature should be raised to double its volume,

a) 566K [Correct]

b) 283K

c) 293K

d) 283 degrees C

Q6. The rate of diffusion of hydrogen (H2) compared with helium (He) is,

a) 0.5 times

b) 1.4 times [Correct]

c) 2 times

d) 4 times

Q7. The non-ideal behaviour results chiefly from

a) Intermolecular attraction and infinite volume

b) Elastic collisions and finite volume

c) Intermolecular attractions and finite volume [Correct]

d) Intermolecular attraction only

Q8. The molar volume of helium (He) is 44.8 dm^3 at,

a) 100 degrees C and 1atm

b) 25 degrees C and 0.25atm

c) 0 degrees C and 0.5atm [Correct]

d) 40 degrees C and 0.5atm

Q9. Which statement about the behaviour of the particles in a gas is not correct

a) They are able to move at great speeds

b) The forces of attraction between the particles are negligible

c) There is large space among the particles

d) They are arranged in regular patterns [Correct]

Q10. At the same temperature and pressure which of the following gases has the greatest density,

a) CO2

b) SO2

c) Cl2 [Correct]

d) H2O

Q11. Weight of one dm^3 of O2 at STP is

a) 1.4384 g [Correct]

b) 1.5394 g

c) 1.6384 g

d) 1.3384 g

Q12. The value of ideal gas constant in dm^3.torr.K^-1.mol^-1

a) 0.0821

b) 1.98722

c) 62.364 [Correct]

d) 8.3143

Q13. 760 torr is equal to,

a) 760 Pascal

b) 76 Pascal

c) 101325 Pascal [Correct]

d) 1.01325 Pascal

Q14. At 50 degrees C a gas has 1 atm pressure, and 20 dm^3 volume, its volume at STP would be

a) 16.94 dm^3 [Correct]

b) 10.92 dm^3

c) 3.66 dm^3

d) 42.2 dm^3

Q15. Which of the following gases will have the fastest effusion rate?

a) CH4 [Correct]

b) NH3

c) CO2

d) O2

Exercise — Short Questions

Q1. Justify that 1 cm^3 of hydrogen (H2) and 1 cm^3 of methane (CH4) at STP will have same number of molecules although one molecule of methane (CH4) is 8 times heavier than that of hydrogen (H2).

By Avogadro's law, equal volumes of gases at the same temperature and pressure contain the same number of molecules, so 1 cm^3 of H2 and 1 cm^3 of CH4 at STP hold equal numbers of molecules. But one CH4 molecule weighs 16 units while one H2 molecule weighs 2 units, that is 8 times as much, so the methane sample is 8 times heavier even though the molecule counts are equal.

Q2. Do you think that some of the postulates of kinetic molecular theory of gases are faulty? Write down these postulates.

Yes, two postulates are faulty for real gases. Kinetic theory assumes gas molecules are point masses with no volume of their own, and that there are no forces of attraction between them. Both fail for real gases: molecules do occupy space, and at close range they do attract one another, which is why real gases deviate from the ideal gas equation.

Q3. Why high pressure and low temperature make the gases non-ideal?

At high pressure the molecules are squeezed close together, so their own volume is no longer negligible beside the volume of the container, and the attractive forces between them become significant. At low temperature the molecules move slowly, so attractions have time to pull them together and reduce the pressure below the ideal value. Both effects make PV differ from nRT.

Q4. What is your opinion about rapid expansion of gases causes cooling?

When a compressed gas expands suddenly into a low-pressure region, it must do work against the attractive forces between its molecules to pull them apart. The energy for this work comes from the kinetic energy of the

molecules themselves, so their average speed falls and the temperature drops. This cooling on free expansion is the Joule-Thomson effect.

Q5. Why do you think that lighter gases can diffuse more rapidly than heavier ones?

At the same temperature all gas molecules have the same average kinetic energy, 1/2 m v^2. A lighter molecule must therefore move faster to have the same kinetic energy as a heavier one. By Graham's law the rate of diffusion is inversely proportional to the square root of the molar mass, so lighter gases diffuse more rapidly than heavier gases.

Exercise — Numerical Questions

Q1. A mass of gas is under a pressure of 760 torr and occupies volume of 525 cm^3. If the pressure is doubled, what volume would the gas now occupy? Assume the temperature is constant. (Answer: 262.5 cm^3)

Step 1: By Boyle's law, P1 V1 = P2 V2 at constant temperature. Step 2: V2 = (760 x 525) / 1520 = 262.5 cm^3. Answer: 262.5 cm^3, as printed in the book.

Q2. Hydrogen gas diffuses through a porous plate at a rate of 500 cm per minute at 0 degrees C. What is the rate of diffusion of oxygen gas through the same porous plate at 0 degrees C? (Answer: 125 cm)

Step 1: By Graham's law, rate of O2 / rate of H2 = sqrt(M of H2 / M of O2) = sqrt(2/32) = 1/4. Step 2: Rate of O2 = 500 / 4 = 125 cm per minute. Answer: 125 cm, as printed in the book.

Q3. It was desired to obtain a volume of 1000 cm^3 of oxygen at 100 degrees C and 640 mm Hg. How many moles of oxygen would be required? (Answer: 0.0275 mol)

Step 1: Convert: P = 640/760 = 0.842 atm, V = 1000 cm^3 = 1 dm^3, T = 100 + 273 = 373 K. Step 2: n = PV / RT = (0.842 x 1) / (0.0821 x 373)

= 0.0275 mol. Answer: 0.0275 mol, as printed in the book.

Q4. Calculate the density of CH4 at 0 degrees C and 1 atmosphere. (Answer: 0.7139 g dm^-3)

Step 1: Density = PM / RT, with M of CH4 = 16 g/mol. Step 2: Density = (1 x 16) / (0.0821 x 273) = 0.714 g/dm^3. Answer: 0.7139 g/dm^3, as printed in the book.

Q5. A sample of Krypton with a volume of 6.25 dm^3 and a pressure of 765 torr and a temperature of 20 degrees C is expanded to a volume of 9.55 dm^3 and a pressure of 375 torr. What will be its final temperature? (Answer: 219.64 K)

Step 1: By the combined gas law, T2 = (P2 V2 T1) / (P1 V1). Step 2: T2 = (375 x 9.55 x 293) / (765 x 6.25) = 219.64 K. Answer: 219.64 K, as printed in the book.

Exercise — Descriptive Questions

Q1(a). Explain the concept of diffusion and effusion of gases. (b) Apply the knowledge of the kinetic theory of gases and derive an expression of Graham's law of diffusion.

Diffusion is the spontaneous mixing of gases by random molecular motion until a uniform mixture forms, while effusion is the escape of gas molecules through a tiny hole into a vacuum or a lower-pressure region without collisions. From kinetic theory, average kinetic energy is (1/2) m v^2 = (3/2) kT, so molecular speed v is proportional to 1/sqrt(m). The rate of diffusion or effusion is proportional to molecular speed, giving Graham's law: rate1/rate2 = sqrt(M2/M1) = sqrt(d2/d1).

Q1(c). The rate of effusion of an unknown gas through a pinhole is found to be 0.279 times the rate of effusion of H2 through the same pinhole. Calculate the molecular mass of the unknown gas at STP. (Answer: 25.7 g)

Step 1: rate(unknown)/rate(H2) = sqrt(M of H2 / M of unknown), so 0.279 = sqrt(2/M). Step 2: Square both sides: 0.0778 = 2/M. Step 3: M = 2 / 0.0778 = 25.7 g/mol. Answer: 25.7 g, as printed in the book.

Q2(a). State Dalton's law of partial pressure. Write down its mathematical form.

Dalton's law states that the total pressure of a mixture of non-reacting gases is equal to the sum of the partial pressures of the individual gases: P(total) = P1 + P2 + P3 + ... Each partial pressure is the pressure the gas would exert alone in the same volume at the same temperature, and Pi = ni RT / V.

Q2(b). Explain the significance of absolute zero, giving its value in degree Celsius and Kelvin.

Absolute zero is the lowest possible temperature, at which the volume of an ideal gas would become zero and molecular motion would be minimum. Its value is -273.15 degrees Celsius, which is taken as 0 Kelvin, the zero of the Kelvin scale. It is significant because it gives a true zero for temperature, so gas laws using absolute temperature work correctly.

Q2(c). What pressure is exerted by a mixture of 2 g of helium (He), 16 g of oxygen (O2) and 10 g of carbon dioxide (CO2) at 10 degrees C in a 5 dm^3 vessel? (Answer: 5.703 atm)

Step 1: Moles: He = 2/4 = 0.5 mol, O2 = 16/32 = 0.5 mol, CO2 = 10/44 = 0.227 mol. Step 2: Total moles = 1.227 mol. Step 3: P = nRT/V = (1.227 x 0.0821 x 283) / 5 = 5.70 atm. Answer: 5.703 atm, as printed in the book.

Q3(a). How would you relate temperature to the average kinetic energy of the particles in a substance?

The average kinetic energy of gas particles is directly proportional to the absolute temperature: average K.E. = (3/2) kT per molecule, or (3/2) RT per mole. Raising the temperature raises the average speed of the

particles, while at absolute zero the average kinetic energy would be zero for an ideal gas.

Q3(b). Explain Linde's method for the liquefaction of gases.

Linde's method liquefies gases by repeated Joule-Thomson cooling. The gas is compressed, cooled to remove the heat of compression, and then allowed to expand through a nozzle into a low-pressure region, where it cools further. The cooled gas is passed back over the incoming gas in a heat exchanger, cooling it still more, and the cycle is repeated until the gas reaches its liquefaction temperature and condenses.

Q3(c). Give two points of evidence to show that gases do not behave ideally under all conditions of temperature and pressure.

Two pieces of evidence are: at high pressure the product PV is not constant as Boyle's law requires, and the compressibility factor Z = PV/nRT differs from 1; and at low temperature gases liquefy, which ideal gases with no attractions never would. Both show that real gases have molecular volume and intermolecular forces that the ideal model ignores.

Q4(a). Write the Van der Waals equation for a real gas. Clearly explain the meaning of the corrective terms for pressure and volume.

The van der Waals equation is (P + a n^2/V^2)(V - nb) = nRT. The term a n^2/V^2 corrects the pressure: attractions between molecules reduce the pressure on the walls, so the observed pressure is increased by this amount to get the ideal pressure. The term nb corrects the volume: the molecules themselves occupy volume, so the free volume is the container volume minus nb.

Q4(b). The temperature of a real gas usually drops when it is allowed to enter into a low pressure (vacuum), Explain.

When a real gas expands into a vacuum or a low-pressure region, its molecules must move apart against their mutual attractions. The work

done against these forces comes from the molecules' own kinetic energy, so the average kinetic energy and therefore the temperature falls. An ideal gas, with no attractions, would show no such cooling.

Q4(c). Define and describe the properties of plasma.

Plasma is the fourth state of matter: a gas in which a large fraction of the atoms are ionised into free electrons and positive ions. It conducts electricity, responds strongly to magnetic fields, and glows when excited. Most of the visible universe is plasma, including the Sun and other stars, lightning, and neon signs.

Q5(a). State basic postulates of kinetic molecular theory of gases.

The basic postulates are: a gas consists of a large number of tiny particles in rapid random motion; the particles are point masses with negligible volume; there are no attractive or repulsive forces between them; collisions are perfectly elastic; the pressure comes from collisions with the walls; and the average kinetic energy is proportional to the absolute temperature.

Q5(b). Explain how Graham's law of diffusion can be derived from the Kinetic Molecular Theory.

From kinetic theory, the average kinetic energy (1/2) m v^2 is the same for all gases at one temperature, so the molecular speed v is proportional to 1/sqrt(m). Since the rate of diffusion is proportional to molecular speed, rate1/rate2 = v1/v2 = sqrt(m2/m1), which is Graham's law of diffusion.

Q5(c). Explain the absolute zero on the basis of Charles' law.

By Charles' law, the volume of a gas at constant pressure is proportional to its absolute temperature, V = kT. A graph of volume against Celsius temperature is a straight line that, extended backwards, meets the temperature axis at -273.15 degrees Celsius, where the volume would be zero. This intercept is absolute zero, the zero of the Kelvin scale.

Project

Project 1. Use the daily weather reports in the newspapers or news and conduct a meteorological research to study the interrelationships among atmospheric pressure (air pressure), temperature, humidity, and other weather variables. Prepare a report of a week explaining your results and present it in the class.

Step 1: Collect the daily weather report from a newspaper or news bulletin for seven days, noting air pressure, temperature and humidity each day. Step 2: Tabulate the three quantities day by day and look for patterns, such as falling pressure with rising humidity before rain. Step 3: Write a one-week report explaining how pressure, temperature and humidity varied together, and present it to the class with your table.

Project 2. The pressure of air in car tyres is checked regularly for safety and to prevent uneven tyre wear and tear. What units of measurement are used on a typical tyre gauge? Also, find out that how gauge pressure relates to atmospheric pressure.

Step 1: Read a tyre gauge and note its units, commonly pounds per square inch (psi), bar or kilopascals (kPa). Step 2: Remember that a gauge reads gauge pressure, which is the pressure above the surrounding atmosphere. Step 3: Absolute pressure = gauge pressure + atmospheric pressure (about 14.7 psi or 1 atm). Step 4: Report your findings, explaining why a tyre that reads zero on the gauge is still full of air at atmospheric pressure.

Unit 5 — States of Matter II: Liquids

Self-Assessment

Q1. Define dynamic equilibrium between two physical states.

Dynamic equilibrium between two physical states means that the two

opposite changes go on at the same time and at equal rates, so that the amounts of the two states stay constant although molecules keep changing state. For example, in a closed bottle, liquid evaporates and vapour condenses at equal rates, so the vapour pressure stays steady.

Q2. Define liquid crystals.

Liquid crystals are substances that flow like liquids but keep some order of a crystal. Their molecules are arranged in a regular pattern in one direction while moving freely in other directions, so they behave partly like a liquid and partly like a solid crystal.

Q3. Differentiate liquid crystals from pure liquids and crystalline solids.

Pure liquids have no regular arrangement of molecules and flow freely in all directions. Crystalline solids have a fully regular three-dimensional arrangement and a fixed shape. Liquid crystals lie between the two: they flow like liquids but their molecules keep a partial order, like crystals, in at least one direction.

Q4. What is energetics of phase change?

The energetics of a phase change is the heat absorbed or released when a substance changes state, such as the molar heat of fusion for melting, the molar heat of vaporisation for boiling, and the molar heat of sublimation for solid to vapour. The heat is used to overcome intermolecular forces, not to raise the temperature.

Q5. Briefly explain the uses of liquid crystals in daily life.

Liquid crystals are used in the displays of watches, calculators, mobile phones, televisions and laptops, where an electric field changes their alignment to form letters and pictures. They are also used in thermometers that change colour with temperature, in medical imaging, and in optical shutters.

Exercise — Choose the Correct Option

Q1. Forces of attraction which may be present between molecules are

a) Intramolecular

b) Intermolecular [Correct]

c) Van der Waal

d) Dipole-induced dipole

Q2. When water freezes at 0 degrees C its density decreases due to

a) Change of bond angles

b) Decrease in volume

c) Empty space present in the structure of ice [Correct]

d) Change of bond length

Q3. London dispersion forces are only forces present among which of the following?

a) Polar molecules of water at room temperature

b) Molecules of HF at room temperature

c) Molecules of solid iodine at room temperature [Correct]

d) Molecules of hydrogen chloride gas at room temperature

Q4. Molar heat of vaporization of water is

a) 40.7 kJ/mole [Correct]

b) 40.7 J/mole

c) 40.7 cal/mole

d) 40.7 kcal/mole

Q5. The S.I unit of viscosity is

a) kg m^-1 s [Correct]

b) kg m s^-1

c) kg m s

d) kg m s^-1

Q6. Liquid gets the shape of the container when it is poured into it. Which

one of the following reasons justifies it?

a) Liquid is incompressible

b) Liquid does not has definite volume

c) Liquid is highly compressible

d) Liquid molecules can slide over each other [Correct]

Q7. Hydrogen bonding is not involved in

a) DNA structure

b) The liquid properties of water

c) Liquid HF

d) Liquid CH4 [Correct]

Q8. Which one is incorrect for evaporation

a) Surface phenomenon

b) Continuous process

c) Exothermic process [Correct]

d) Causes cooling

Q9. Exceptionally low acidic strength of HF is due to

a) strong polar bond

b) small size of fluorine

c) strong hydrogen bonding [Correct]

d) Van der Waals forces

Q10. Distillation under very reduced pressure is

a) Vacuum distillation [Correct]

b) Steam distillation

c) Destructive Distillation

d) Fractional distillation

Q11. Which one of the following is more temperature sensitive

a) Ionic crystals

b) Solid crystals

c) Liquid crystals [Correct]

d) Molecular crystals

Q12. Which of the following has weakest London dispersion forces

a) F2 [Correct]

b) Cl2

c) Br2

d) I2

Q13. What is the boiling point of water at a mountain,

a) It is 100 degrees C

b) It is < 100 degrees C, since the atmospheric pressure is greater

c) It is > 100 degrees C, since the atmospheric pressure is greater

d) It is < 100 degrees C, since the atmospheric pressure is less. [Correct]

Exercise — Short Questions

Q1. Give at least two of the effects on our lives if water had weak hydrogen bonding among its molecules.

Two effects would follow at once. First, water would boil at a much lower temperature and evaporate very quickly, so rivers, lakes and our bodies would lose water fast. Second, ice would no longer float and aquatic life would be disturbed, because the open hydrogen-bonded structure of ice and the high heat capacity of water both depend on strong hydrogen bonding.

Q2. Give the reason that why HF is a liquid at ordinary temperature while HCl is a gas.

HF molecules form strong hydrogen bonds with one another, holding them together as a liquid at ordinary temperature. HCl molecules have only weak dipole-dipole forces, which are easily overcome by thermal motion, so HCl escapes as a gas at ordinary temperature.

Q3. Why H2O has high boiling point than HF, although fluorine is more electronegative than oxygen.

Each water molecule can form up to four hydrogen bonds, two through its hydrogen atoms and two through the lone pairs on oxygen, building a wide network. Each HF molecule can form only about two hydrogen bonds on average, so the network in water is stronger and more energy is needed to break it, giving water the higher boiling point despite fluorine being more electronegative.

Q4. How would you justify that water and ethanol can mix easily in all proportions?

Both water and ethanol form hydrogen bonds, and the -OH group of ethanol fits into the hydrogen-bonded network of water. The attraction between water and ethanol molecules is therefore as strong as the attraction within each liquid, so they mix in all proportions without separating.

Q5. How can you explain that neon and argon both are mono-atomic noble gases of the same group, neon has boiling of -248 degrees C, while argon has -189 degrees C.

Both are mono-atomic noble gases, but argon atoms are larger and have more electrons than neon atoms, so the London dispersion forces between argon atoms are stronger. More energy is therefore needed to separate argon atoms, giving argon a higher boiling point (-189 degrees C) than neon (-248 degrees C).

Q6. How would you justify that different liquids have different rates of evaporation even at the same temperature.

Different liquids have different strengths of intermolecular forces. A liquid with weak forces, like ether, lets its surface molecules escape easily, so it evaporates fast at a given temperature. A liquid with strong hydrogen

bonding, like water, holds its molecules back, so it evaporates slowly at the same temperature.

Q7. Justify the statement that earthenware vessels keep water cool even in hot summer days.

An earthenware vessel is porous, so a little water seeps through its walls and evaporates from the outer surface. Evaporation needs latent heat, which it takes from the water inside, cooling it. The vessel therefore keeps water cool even on hot summer days.

Q8. How vacuum distillation can be used to avoid decomposition of sensitive liquids?

Lowering the pressure lowers the boiling point of a liquid, because a liquid boils when its vapour pressure equals the external pressure. A heat-sensitive liquid can therefore be distilled at a low temperature under vacuum, below the temperature at which it would decompose at normal pressure.

Q9. Why evaporation of a liquid causes cooling?

Evaporation takes the fastest molecules from the surface of a liquid, and those molecules carry away kinetic energy. The average kinetic energy of the molecules left behind falls, and since temperature measures average kinetic energy, the liquid cools. This is why evaporation causes cooling.

Q10. Explain that why a liquid boils at different temperature at sea level and mountains.

A liquid boils when its vapour pressure equals the external atmospheric pressure. At sea level the atmospheric pressure is 1 atm, so water boils at 100 degrees C. On a mountain the atmospheric pressure is lower, so the vapour pressure reaches it at a lower temperature, and water boils below 100 degrees C.

Q11. Why water droplet is spherical?

Surface tension pulls the surface of a liquid into the smallest possible area for its volume, and the shape with the smallest surface area for a given volume is a sphere. A falling droplet therefore becomes spherical to minimise its surface energy.

Q12. Temperature of a liquid remains constant during boiling although heat is being supplied continuously, why?

During boiling the heat supplied is not used to raise the temperature but to overcome the intermolecular forces and change the liquid into vapour, as latent heat of vaporisation. The temperature stays constant until all the liquid has boiled, even though heat is being supplied continuously.

Exercise — Descriptive Questions

Q1(a). Briefly explain the properties of liquids (i) Diffusion, (ii) Compression, (iii) Expansion, (iv) Motion of Molecules, (v) Intermolecular Forces on the basis of Kinetic Molecular Theory.

On the basis of kinetic molecular theory: (i) Diffusion: liquid molecules are in constant motion and can move from one place to another, so two liquids mix slowly by diffusion. (ii) Compression: liquid molecules are already close together, so liquids are almost incompressible. (iii) Expansion: on heating, molecular motion increases and the molecules move slightly apart, so liquids expand a little. (iv) Motion of molecules: molecules slide over one another but cannot move freely like gas molecules. (v) Intermolecular forces: forces between liquid molecules are strong enough to hold them together but weak enough to let them flow.

Q1(b). What is anomalous behaviour of water when its density shows maximum at 4 degrees C?

Water is unusual because its density increases as it cools down to 4 degrees C and then decreases as it cools further to 0 degrees C, so water

has maximum density at 4 degrees C. Below 4 degrees C the molecules begin to form the open hydrogen-bonded structure of ice, which takes up more space, so ice is less dense than water and floats.

Q1(c). Wet clothes dry more quickly on a hot, dry day than on a hot, humid day. Explain.

On a hot, dry day the air holds little water vapour, so evaporation from wet clothes is fast and the clothes dry quickly. On a hot, humid day the air is already nearly saturated with water vapour, so evaporation is slow and the clothes dry slowly, even though the temperature is high.

Q2(a). Define and explain these terms along with their units: (i) Molar Heat of Vaporization, (ii) Molar heat of Fusion, (iii) Molar Heat of Sublimation.

Molar heat of vaporisation is the heat absorbed when one mole of a liquid changes into vapour at its boiling point, in kJ/mol; for water it is 40.7 kJ/mol. Molar heat of fusion is the heat absorbed when one mole of a solid melts at its melting point, in kJ/mol. Molar heat of sublimation is the heat absorbed when one mole of a solid changes directly into vapour, in kJ/mol, and it equals the sum of the heats of fusion and vaporisation.

Q2(b). Explain the origin of the London Force that exists between molecules.

London forces arise from temporary, instantaneous dipoles. At any instant the electrons in an atom may be unevenly distributed, creating a tiny temporary dipole that induces a dipole in a neighbouring atom, and the two attract. These forces exist between all molecules, grow stronger with more electrons and larger atoms, and are the only forces between non-polar molecules.

Q3(a). Define boiling point and how does the boiling point of a liquid dependent on the external atmospheric pressure?

Boiling point is the temperature at which the vapour pressure of a liquid becomes equal to the external pressure. Because of this, the boiling point depends on the external pressure: at higher pressure, as in a pressure cooker, the boiling point rises; at lower pressure, as on a mountain, the boiling point falls. The normal boiling point is measured at 1 atm pressure.

Q3(b). Briefly explain physical properties of water with special reference to (i) Evaporation (ii) Vapour Pressure (iii) Boiling Point (iv) Viscosity (v) Surface Tension

Water evaporates readily at all temperatures and its vapour pressure rises with temperature; it boils at 100 degrees C at 1 atm because of strong hydrogen bonding. Its viscosity is low but higher than that of non-polar liquids, and its surface tension is high (72.8 mN/m) because surface molecules are pulled inwards by hydrogen bonds. These properties all follow from the strong hydrogen-bonded network of water.

Q3(c). Why does the vapour pressure of a liquid depend on the intermolecular forces?

Vapour pressure depends on intermolecular forces because only molecules with enough kinetic energy to overcome the attractions of their neighbours can escape into the vapour. In a liquid with weak forces many molecules escape, giving a high vapour pressure; in a liquid with strong hydrogen bonding few escape, giving a low vapour pressure.

Q4(a). What is the relationship between the intermolecular forces that exist in a liquid and its surface tension?

Surface tension is the result of the intermolecular forces pulling surface molecules inwards, since they have no neighbours above them. The stronger the intermolecular forces, the greater the inward pull and the higher the surface tension. Water, with strong hydrogen bonding, has a

high surface tension, while ether, with weak forces, has a low one.

Q4(b). What do you know about dipole-dipole interactions?

Dipole-dipole interactions are the attractions between the positive end of one polar molecule and the negative end of another polar molecule. They are stronger than London forces but weaker than hydrogen bonds, and they raise the boiling points of polar liquids such as HCl and acetone above those of comparable non-polar liquids.

Q5(a). Differentiate liquid crystals from pure liquids and crystalline solids? Also give some uses of liquid crystals in daily life.

Liquid crystals flow like liquids but keep a partial order like crystals, while pure liquids have no order at all and crystalline solids have full order and a fixed shape. Liquid crystals are used in LCD screens of watches, calculators, phones and televisions, in temperature-sensitive thermometers, and in optical devices that respond to electric fields.

Q5(b). Define phase change. Name all possible changes that can occur among the vapour, liquid and solid states of a substance.

A phase change is a change of a substance from one physical state to another, such as melting, boiling or freezing. The possible changes among vapour, liquid and solid are: solid to liquid (melting), liquid to solid (freezing), liquid to vapour (vaporisation), vapour to liquid (condensation), solid to vapour (sublimation) and vapour to solid (deposition or reverse sublimation).

Project

Project 1. Develop a set of criteria that will allow you to classify the following substances on the basis of intermolecular force i.e. Dipole-dipole forces, Hydrogen bonding and London dispersion forces: Water, Milk, Honey, Methylated Sprit, Glycerine ($C_3H_8O_3$), Carbon Tetrachloride

(CCl4), Acetone (CH3COCH3), Chloroform (CHCl3), Benzene (C6H6), Kerosene oil, Ammonia (NH3), Hydrochloric Acid (HCl) and Methane (CH4). Present it in the classroom and give reason of your criteria of classification based on their molecular structures.

Step 1: Sort the substances by their strongest intermolecular force: hydrogen bonding for water, milk, honey, methylated spirit, glycerine and ammonia; dipole-dipole for acetone, chloroform and hydrogen chloride; London dispersion only for carbon tetrachloride, benzene, kerosene oil and methane. Step 2: Present the table in class and explain each placement from the molecular structure, for example the O-H groups in water and glycerine give hydrogen bonding, while the symmetrical CCl4 and C6H6 molecules are non-polar. Step 3: Conclude that the force type follows from polarity and the presence of H attached to N, O or F.

Project 2. Classify the above given substances on the basis of more volatile, less volatile and non-volatile at the same temperature based upon intermolecular forces, with suitable reason.

Step 1: List the substances from most to least volatile at the same temperature: methane and kerosene fractions first, then acetone, chloroform and carbon tetrachloride, then methylated spirit and ammonia, and last water, milk, honey and glycerine. Step 2: Explain that volatility falls as intermolecular forces strengthen: London-only liquids escape easily, dipole liquids less easily, and hydrogen-bonded liquids least easily. Step 3: Present the ranking with reasons, noting that honey and glycerine are least volatile because of their extensive hydrogen bonding and high viscosity.

Unit 6 — States of Matter III: Solids

Self-Assessment

Q1. Explain the significance of the unit cell to the shape of the crystal using NaCl as an example.

The unit cell is the smallest repeating unit of a crystal; repeating it in three dimensions builds the whole crystal and decides its shape. In NaCl the unit cell is face-centred cubic: the whole crystal grows by repeating this cubic cell, so the NaCl crystal itself is cubic in shape. The shape of the crystal is therefore the shape of its unit cell repeated.

Q2. Name three types of packing arrangements and draw or construct models of them.

The three packing arrangements are simple cubic packing, body-centred cubic packing and face-centred cubic (cubic close) packing. In simple cubic packing each sphere touches six neighbours; in body-centred packing each sphere touches eight; in face-centred packing each sphere touches twelve. Models can be built with equal balls, for example marbles or table-tennis balls, stacked in each pattern.

Q3. Name three factors that affect the shape of an ionic crystal.

Three factors are the relative sizes (radius ratio) of the cation and anion, the charges on the ions, and the conditions of temperature and pressure during crystallisation. A large radius ratio favours a high coordination number and a different lattice, while temperature can change one form into another at the transition temperature.

Q4. Differentiate between ionic, covalent, molecular and metallic crystalline solids.

Ionic solids are built from ions held by strong electrostatic forces, so they are hard with high melting points and conduct only when molten or dissolved. Covalent solids have atoms joined by covalent bonds throughout, so they are very hard with very high melting points, like diamond. Molecular solids have molecules held by weak van der Waals

forces, so they are soft with low melting points. Metallic solids have positive ions in a sea of free electrons, so they conduct heat and electricity and are malleable.

Q5. Explain the low density and high heat of fusion of ice.

Ice has a low density because its water molecules are held by hydrogen bonds in an open cage-like structure with large empty spaces, so ice occupies more volume than the same mass of water and floats. Ice has a high heat of fusion because melting must break a large number of hydrogen bonds, which needs a great deal of energy, 6.0 kJ per mole.

Q6. Define and explain molecular and metallic solids.

Molecular solids consist of atoms or molecules held together by weak van der Waals forces or hydrogen bonds, so they are soft with low melting points, like ice, iodine and sugar. Metallic solids consist of positive metal ions surrounded by a sea of delocalised valence electrons, so they are hard, conduct heat and electricity, and can be hammered and drawn into wires, like copper and iron.

Exercise — Choose the Correct Option

Q1. The crystal of ice in an example of

a) Metallic solid

b) Molecular solid [Correct]

c) Covalent solid

d) Ionic solid

Q2. KNO3 and NaNO3, both have the atomic ratio of 1:1:3 in their crystals, property is

a) Isotropy

b) Anisotropy

c) Isomorphism [Correct]

d) Polymorphism

Q3. The number of center of symmetry of a cubic crystal is/are

a) One [Correct]

b) Two

c) Three

d) Six

Q4. A pair of compounds, which are isomorphic in nature is,

a) NaCl and KNO_3

b) MgO and ZnO [Correct]

c) $NaNO_3$ and CdS

d) NaF and $CaCO_3$

Q5. Crystals show variation in physical properties depending upon the direction. The property is called

a) Isomorphism

b) Polymorphism

c) Anisotropy [Correct]

d) Isotropy

Q6. The parameters of unit cell is

a) 4

b) 5

c) 6 [Correct]

d) 7

Q7. Sodium chloride (NaCl) on crystallization form the

a) Face centered cubic lattice [Correct]

b) Orthorhombic lattice

c) Body centered cubic lattice

d) Simple cubic lattice

Q8. The molecules of CO_2 in dry ice form the

a) Ionic crystals

b) Covalent crystals

c) Molecular crystals [Correct]

d) Metallic crystal

Q9. The nature of bonds between atoms in graphite is

a) Ionic

b) Covalent [Correct]

c) Molecular

d) Metallic

Q10. The example of metallic solid is

a) C

b) B

c) Si

d) Cu [Correct]

Q11. In solid state, the crystals which are good conductor of heat and electricity

a) Covalent

b) ionic

c) Molecular

d) Metallic [Correct]

Q12. When a crystalline substance conducts current in one direction but not through other directions of the crystal, this property is

a) Anisotropy [Correct]

b) Isotropy

c) Allotropy

d) Isomorphism

Q13. Crystals of the same substance can co-exist in equilibrium with each other at its

a) Boiling point

b) Melting point

c) Transition temperature [Correct]

d) Heat of fusion

Q14. Crystal lattice is an arrangement of particles in

a) Four dimensions

b) Three dimensions [Correct]

c) Two dimensions

d) One dimension

Exercise — Short Questions

Q1. Briefly explains the term unit cell and crystal lattice.

A crystal lattice is the regular three-dimensional arrangement of points that represent the positions of particles in a crystal. A unit cell is the smallest portion of that lattice which, repeated in all directions, reproduces the whole crystal; it is defined by its edge lengths and angles.

Q2. Why crystalline solids are anisotropic?

In a crystalline solid the particles are arranged regularly, so the distances and forces between particles differ in different directions. A property such as electrical conductivity, refractive index or cleavage therefore has different values when measured along different directions of the crystal, which is anisotropy.

Q3. Give the reason that metallic crystals are good conductors of electricity but ionic crystals are not.

In a metallic crystal the valence electrons are free to move through the whole crystal as a sea of electrons, so an electric current flows easily. In an ionic crystal the ions are fixed in their lattice positions in the solid state and cannot move, so solid ionic crystals do not conduct; they

conduct only when molten or dissolved, when the ions become free.

Q4. Differentiate between crystalline and amorphous solids.

Crystalline solids have a regular arrangement of particles, a definite geometrical shape, sharp melting points, cleavage planes and anisotropy. Amorphous solids have no regular arrangement, no definite shape, soften over a range of temperature instead of melting sharply, and show no cleavage planes or anisotropy, like glass and plastic.

Q5. Justify that crystalline solids have sharp melting point.

All the particles in a crystalline solid are held by identical forces in a regular lattice, so at one fixed temperature all the bonds break together and the whole solid melts at once. The melting point is therefore sharp. In an amorphous solid the forces differ from place to place, so it softens gradually over a range of temperature.

Q6. What is transition temperature? Give examples.

Transition temperature is the temperature at which one crystalline form of a substance changes into another form, both forms existing together in equilibrium at that temperature. Examples are sulphur, rhombic changing to monoclinic at 95.6 degrees C, and tin, grey changing to white at 13.2 degrees C.

Q7. Explain that isomorphic substances have usually the same atomic ratio.

Isomorphic substances have the same crystal shape, which requires the same arrangement of particles in the unit cell, and that arrangement is possible only when the substances have the same atomic ratio. For example, KNO3 and NaNO3 both have the atomic ratio 1:1:3 and the same crystal form, which is why isomorphism usually goes with the same atomic ratio.

Q8. Give the reason that covalent crystals are hard while molecular

crystals are soft.

In a covalent crystal every atom is joined to its neighbours by strong covalent bonds running through the whole crystal, as in diamond, so the crystal is very hard. In a molecular crystal the molecules are held only by weak van der Waals forces, which break easily, so the crystal is soft and melts at a low temperature.

Q9. Define lattice energy and give examples.

Lattice energy is the energy released when one mole of an ionic crystal is formed from its gaseous ions of opposite charge. For example, the lattice energy of NaCl is about 787 kJ/mol and of MgO about 3795 kJ/mol, the higher value for MgO reflecting the doubly charged ions.

Q10. Differentiate between polymorphism and isomorphism.

Polymorphism is the occurrence of the same substance in more than one crystalline form, such as rhombic and monoclinic sulphur. Isomorphism is the occurrence of different substances in the same crystalline form, such as NaNO3 and CaCO3. Polymorphism concerns one substance in many forms; isomorphism concerns many substances in one form.

Exercise — Descriptive Questions

Q1(a). Explain the kinetic molecular interpretation of solids.

In a solid the particles are closely packed and cannot move freely; they only vibrate about their mean positions, so kinetic energy exists as vibrational energy. The close packing gives solids a definite shape and volume, minimum empty space, negligible diffusion and almost no compressibility.

Q1(b). Discuss in detail the following properties, keeping in view the Kinetic Molecular Theory. (i) Diffusion (ii) Compression (iii) Expansion (iv) Motion of molecules

On the kinetic molecular view: (i) Diffusion is negligible because particles cannot leave their positions. (ii) Compression is negligible because particles are already in contact. (iii) Expansion is small because heating only increases the amplitude of vibration. (iv) Motion of molecules is vibration about fixed mean positions, not free movement.

Q2(a). Discuss in detail a sodium chloride lattice.

Sodium chloride forms a face-centred cubic lattice. Each Na+ ion is surrounded by six Cl- ions and each Cl- ion by six Na+ ions, so the coordination number is 6:6. The unit cell contains four Na+ and four Cl- ions, that is four formula units of NaCl. The ions are arranged so that the crystal cleaves along planes and grows as cubes.

Q2(b). Describe the following properties of crystalline solids: (i) Habit of a crystal (ii) Cleavage planes (iii) Crystal growth (iv) Anisotropy (v) Symmetry.

Habit of a crystal is the shape in which it usually grows, such as the cubes of NaCl. Cleavage planes are the planes along which a crystal splits cleanly when struck, because the bonds are weakest there. Crystal growth proceeds by adding layers of particles to the faces of a seed crystal. Anisotropy means properties vary with direction in the crystal. Symmetry means the crystal looks unchanged after rotation, reflection or inversion through a centre of symmetry.

Q3. Differentiate between the following pairs: (i) Ionic solid and molecular solid (ii) Lattice energy and crystal lattice (iii) Plane of symmetry and axis of symmetry

An ionic solid is built from ions held by strong electrostatic forces, so it is hard with a high melting point and conducts only when molten or dissolved; a molecular solid is built from molecules held by weak forces, so it is soft with a low melting point and does not conduct. Lattice energy

is the energy released in forming a crystal from gaseous ions, while a crystal lattice is the arrangement of the particles themselves. A plane of symmetry divides a crystal into two mirror halves, while an axis of symmetry is a line about which rotation leaves the crystal looking unchanged.

Q4(a). Define allotropy. Use oxygen and sulphur as an example to explain allotropes.

Allotropy is the occurrence of an element in more than one form in the same physical state. Oxygen exists as O2 and as ozone O3, which differ in properties although both are oxygen. Sulphur exists as rhombic sulphur and monoclinic sulphur, which have different crystal shapes and change into each other at 95.6 degrees C.

Q4(b). Explain the low density and high heat of fusion of ice.

Ice floats because hydrogen bonding holds its molecules in an open structure with large empty spaces, giving ice a lower density than water. Ice has a high heat of fusion, 6.0 kJ/mol, because melting it requires breaking a large number of hydrogen bonds, which absorbs a great deal of energy without raising the temperature.

Q5(a). Define and explain molecular and metallic solids

Molecular solids are made of molecules held by weak van der Waals forces or hydrogen bonds, so they are soft, have low melting points and do not conduct electricity, like ice and iodine. Metallic solids are made of positive ions in a sea of free electrons, so they are hard, have high melting points, conduct heat and electricity, and are malleable and ductile, like copper and iron.

Q5(b). What are the features of three types of packing arrangements in a system and construct models of them?

The three packing arrangements are simple cubic packing, in which each

sphere touches six neighbours and much space is wasted; body-centred cubic packing, in which each sphere touches eight neighbours; and face-centred cubic or close packing, in which each sphere touches twelve neighbours and the packing is most efficient. Models are built by stacking equal balls, such as marbles, in each pattern and comparing the empty space left.

Q6(a). Write down the three factors that affect the shape of an ionic crystal

The three factors are the radius ratio of the cation to the anion, which decides the coordination number; the charges on the ions, which decide the strength of the lattice; and the temperature and pressure of crystallisation, which can favour one form over another.

Q6(b). What is meant by unit cell of crystal? Sketch the unit cell of NaCl.

The unit cell of a crystal is the smallest repeating portion of its lattice that shows the full pattern of the crystal. In NaCl the unit cell is a cube with Cl- ions at the corners and face centres and Na+ ions at the edge centres and the body centre, giving four formula units per cell. A sketch shows this cubic cell with the two kinds of ions in their positions.

Project

Project. Collect the different household items and classify them as amorphous or crystalline substances: Table salt, Sugar, Washing soda, Baking soda, Ice, Glass.

Step 1: Collect household items: table salt, sugar, washing soda, baking soda, ice and glass. Step 2: Test each for a definite shape, a sharp melting point and cleavage: salt, sugar, washing soda, baking soda and ice are crystalline, with regular shapes and sharp melting points. Step 3: Glass is amorphous, with no regular shape, no sharp melting point and no

cleavage planes. Step 4: Present the classification in a table with the test results for each item.

Unit 7 — Chemical Equilibrium

Reading Check

Q1. Define common ion effect.

Common ion effect is the suppression of the ionisation of a weak electrolyte when a strong electrolyte containing a common ion is added to its solution. For example, adding NH4Cl to NH4OH solution increases NH4+ ions, so the equilibrium NH4OH ⇌ NH4+ + OH- shifts backwards and the ionisation of NH4OH is suppressed.

Q2. Explain the solubility product of a substance with an example.

The solubility product (Ksp) of a sparingly soluble salt is the product of the molar concentrations of its ions in a saturated solution, each raised to the power of its coefficient. For example, for AgCl(s) ⇌ Ag+(aq) + Cl-(aq), Ksp = [Ag+][Cl-] = 1.8 x 10^-10 at 25 degrees C. If the ionic product exceeds Ksp, the salt precipitates; if it is less, more salt dissolves.

Exercise — Choose the Correct Answer

Q1. In which of the following values of Kc, the reaction goes to completion in the forward direction:

a) 10^2

b) 10^30 [Correct]

c) 10^-30

d) 1

Q2. An appreciable amount of the product and reactants is present in a reaction if its equilibrium constant value is:

a) Negative and large

b) Negative and small

c) Zero

d) Neither large nor small. [Correct]

Q3. Consider N2(g) + 3H2(g) ⇌ 2NH3(g), delta H = -92.46 kJ/mol. The optimum temperature for the production of ammonia is:

a) 5000 degrees C

b) 450 degrees C [Correct]

c) 0 degrees C

d) Constant temperature

Q4. In the production of SO3 from SO2 and oxygen. The yield of SO3 is increased by

a) increasing temperature

b) adding a catalyst

c) adding more SO2 [Correct]

d) removing O2

Q5. Le-Chatelier's principle applies to

a) mechanical system

b) physical system

c) chemical system

d) both b and c [Correct]

Q6. The unit of Kc for the system PCl5 ⇌ PCl3 + Cl2 is

a) mol^2/dm^6

b) dm^3/mol

c) mol/dm^6

d) mol/dm^3 [Correct]

Q7. Forward reaction goes virtually to completion when Kc is

a) positive and small

b) positive and large [Correct]

c) negative and large

d) unity

Q8. Endothermic reactions are favoured in the forward direction by

a) cooling

b) freezing

c) heating [Correct]

d) adding a catalyst

Q9. Molecules of chlorine decompose to a very small extent into atomic chlorine i.e. $Cl_2 \rightleftharpoons 2Cl$; this is because Kc of this reaction is

a) very large

b) very small [Correct]

c) zero

d) 1

Q10. How much reaction is complete when Kc = 1 for the system $A + B \rightleftharpoons C + D$

a) 10%

b) 25%

c) 50% [Correct]

d) 100%

Q11. HCl when added to H_2S solution

a) suppresses the ionization of H_2S [Correct]

b) enhances the ionization

c) solution becomes coloured

d) does not affect

Q12. The value of Kp is greater than Kc when the difference of the moles of the products and reactants is

a) zero

b) positive [Correct]

c) negative

d) one

Exercise — Short Questions

Q1. The change in temperature changes the equilibrium position of the reaction N2(g) + O2(g) ⇌ 2NO(g) but change in pressure does not, why?

The reaction N2 + O2 ⇌ 2NO has the same number of moles on both sides (2 = 2), so a change in pressure cannot shift the equilibrium in either direction. Temperature changes the value of Kc itself: since the forward reaction is endothermic, heating favours the forward reaction and more NO is formed, while cooling favours the reverse.

Q2. Give the units of Kp for the following reversible reactions. (a) PCl5(g) ⇌ PCl3(g) + Cl2(g) (b) N2(g) + 3H2(g) ⇌ 2NH3(g) (c) H2(g) + CO2(g) ⇌ CO(g) + H2O(g) (d) 4NH3(g) + 5O2(g) ⇌ 4NO(g) + 6H2O(g)

The units of Kp follow delta n, the change in moles of gas. (a) PCl5 ⇌ PCl3 + Cl2, delta n = +1, so Kp has units of atm. (b) N2 + 3H2 ⇌ 2NH3, delta n = -2, so Kp has units of atm^-2. (c) H2 + CO2 ⇌ CO + H2O, delta n = 0, so Kp has no units. (d) 4NH3 + 5O2 ⇌ 4NO + 6H2O, delta n = +1, so Kp has units of atm.

Q3. Why does the value of Kc fall with the rise in temperature for the synthesis of SO3? 2SO2(g) + O2(g) ⇌ 2SO3(g), delta H = -94.58 kJ/mol

The synthesis 2SO2 + O2 ⇌ 2SO3 is exothermic, delta H = -94.58 kJ/mol. Raising the temperature favours the endothermic reverse reaction by Le-Chatelier's principle, so less SO3 is formed at equilibrium and the equilibrium constant Kc becomes smaller as temperature rises.

Q4. There is a dynamic rather than static equilibrium present between liquid and vapour at constant temperature; explain.

In a closed vessel at constant temperature, liquid evaporates and vapour

condenses at the same time. When the two rates become equal, the amount of liquid and vapour stays constant, but molecules keep passing both ways across the surface. The equilibrium is therefore dynamic, not static: nothing appears to change, yet both processes continue.

Q5. The change in concentration of reactants does not change the value of equilibrium constant permanently, elaborate.

Adding a reactant or removing a product disturbs the equilibrium, and the system shifts to restore the ratio of products to reactants given by Kc. Once the new equilibrium is reached, the ratio is the same as before, so Kc itself is unchanged. Only a change in temperature changes Kc permanently, because Kc is a constant at a given temperature.

Q6. Discuss the equilibrium of a sparingly soluble salt.

A sparingly soluble salt in contact with water sets up the equilibrium: solid salt ⇌ its ions in solution. Only a very small amount dissolves, and the solution soon becomes saturated, so the rate of dissolving equals the rate of crystallisation. The equilibrium is described by the solubility product, Ksp, the product of the ion concentrations in the saturated solution.

Q7. Common ion effect operates best in the purification of certain substances, explain.

Common ion effect is used to purify substances by precipitation. For example, pure NaCl is obtained by passing HCl gas through a saturated solution of impure NaCl: the Cl- ion is common to both, so the equilibrium NaCl(s) ⇌ Na+ + Cl- shifts backwards and pure NaCl precipitates, leaving soluble impurities in solution.

Q8. The ionization of calcium oxalate is suppressed by adding CaCl2 to it, why?

Calcium oxalate ionises as CaC2O4 ⇌ Ca2+ + C2O4^2-. Adding CaCl2 adds Ca2+ ions, which are common to the salt, so by Le-Chatelier's

principle the equilibrium shifts backwards to keep Ksp constant. The ionisation of calcium oxalate is therefore suppressed and its solubility falls.

Q9. The solubility of a sparingly soluble substance is calculated from the solubility product data, explain.

For a sparingly soluble salt AB, the solubility product is Ksp = [A+][B-] = s x s = s^2, where s is the solubility in mol/dm^3, so s = sqrt(Ksp). For AgCl with Ksp = 1.8 x 10^-10, s = sqrt(1.8 x 10^-10) = 1.34 x 10^-5 mol/dm^3. For a salt like PbCl2, Ksp = 4s^3, so s = (Ksp/4)^(1/3).

Exercise — Numerical Questions

Q1. Calculate Kc for the synthesis of ammonia at 500 degrees C. Kp = 1.44 x 10^-5.

Step 1: For N2 + 3H2 ⇌ 2NH3, delta n = 2 - 4 = -2. Step 2: Kc = Kp (RT)^-delta n = Kp (RT)^2. Step 3: T = 500 + 273 = 773 K, RT = 0.0821 x 773 = 63.46. Step 4: Kc = 1.44 x 10^-5 x (63.46)^2 = 1.44 x 10^-5 x 4027 = 5.8 x 10^-2. Answer: 5.8 x 10^-2 (worked from the given Kp; the printed answer line at the foot of book p232 is cut in the scan).

Q2. The solubility of CaF2 in water at 25 degrees C is found to be 2.05 x 10^-4 M. What is the Ksp at this temperature? (Ans: 3.446 x 10^-11)

Step 1: CaF2 ⇌ Ca2+ + 2F-, so if solubility is s, [Ca2+] = s and [F-] = 2s. Step 2: Ksp = [Ca2+][F-]^2 = s x (2s)^2 = 4s^3. Step 3: Ksp = 4 x (2.05 x 10^-4)^3 = 3.446 x 10^-11. Answer as printed in the book.

Q3. The solubility product of AgCl is 1.8 x 10^-10. Calculate the molar solubility of the salt. (Ans: 1.34 x 10^-5 M)

Step 1: AgCl ⇌ Ag+ + Cl-, so Ksp = s x s = s^2, where s is the molar solubility. Step 2: s = sqrt(Ksp) = sqrt(1.8 x 10^-10) = 1.34 x 10^-5 M. Answer as printed in the book.

Q4. A mixture of 0.5 moles of H2 and I2 each was placed in a one litre flask at 400 degrees C. Calculate the concentration of H2, I2 and HI at equilibrium. The Kc for the reaction is 54.3. (Ans: [H2] = [I2] = 0.107 M, [HI] = 0.786 M)

Step 1: H2 + I2 ⇌ 2HI. Let x mol react: [H2] = [I2] = 0.5 - x, [HI] = 2x. Step 2: Kc = [HI]^2 / ([H2][I2]) = (2x)^2 / (0.5 - x)^2 = 54.3. Step 3: Taking square roots, 2x / (0.5 - x) = 7.369, so 2x = 3.6845 - 7.369x, giving x = 0.3933. Step 4: [H2] = [I2] = 0.5 - 0.3933 = 0.107 M, [HI] = 0.786 M. Answers as printed in the book.

Q5. The equilibrium mixture contains 1 mole of PCl5, 0.3 mole of PCl3 and 0.08 mole of Cl2 in a 10 L flask. Calculate Kc. (Ans: Kc = 2.40 x 10^-3)

Step 1: Concentrations: [PCl5] = 1/10 = 0.1 M, [PCl3] = 0.3/10 = 0.03 M, [Cl2] = 0.08/10 = 0.008 M. Step 2: Kc = [PCl3][Cl2] / [PCl5] = (0.03 x 0.008) / 0.1 = 2.40 x 10^-3. Answer as printed in the book.

Q6. The equilibrium concentration of N2 and O2 is 0.25 M each while its Kc is 0.1 at 2000 degrees C. Calculate the equilibrium concentration of NO according to the following reaction. N2(g) + O2(g) ⇌ 2NO(g) (Ans: [NO] = 0.079 M)

Step 1: Kc = [NO]^2 / ([N2][O2]) = 0.1. Step 2: [NO]^2 = 0.1 x 0.25 x 0.25 = 0.00625. Step 3: [NO] = sqrt(0.00625) = 0.079 M. Answer as printed in the book.

Exercise - Descriptive Questions

Q1. Write down Kc for the following reversible reactions. i) PCl5(g) ⇌ PCl3(g) + Cl2(g) ii) 2HI(g) ⇌ H2(g) + I2(g) iii) H2(g) + CO2(g) ⇌ CO(g) + H2O(g) iv) 2N2O5(g) ⇌ 4NO2(g) + O2(g)

(i) Kc = [PCl3][Cl2] / [PCl5]. (ii) Kc = [H2][I2] / [HI]^2. (iii) Kc = [CO][H2O] / ([H2][CO2]). (iv) Kc = [NO2]^4 [O2] / [N2O5]^2. In each case the

products are on top and the reactants below, each raised to its coefficient.

Q2. Predict the effect of increasing pressure on the following gaseous equilibria. i) N2(g) + O2(g) ⇌ 2NO(g) ii) PCl5(g) ⇌ PCl3(g) + Cl2(g) iii) 2NO2(g) ⇌ N2O4(g) iv) 2SO2(g) + O2(g) ⇌ 2SO3(g)

(i) No effect: both sides have 2 moles of gas, so pressure cannot favour either side. (ii) Shifts backwards: the forward side has 2 moles against 1 mole of reactant, so higher pressure favours the side with fewer moles, the reactants. (iii) Shifts forwards: 2 moles become 1 mole, so higher pressure favours the product. (iv) Shifts forwards: 3 moles become 2 moles, so higher pressure favours the product.

Q3. Derive the relationship between Kp and Kc.

Step 1: For aA + bB ⇌ cC + dD, Kp = (pC^c pD^d) / (pA^a pB^b) and Kc = ([C]^c [D]^d) / ([A]^a [B]^b). Step 2: From the ideal gas law, p = [gas] RT, so each partial pressure is replaced by concentration times RT. Step 3: Kp = Kc (RT)^((c+d)-(a+b)) = Kc (RT)^delta n, where delta n is the change in moles of gas. If delta n = 0, Kp = Kc.

Q4. What is the solubility product? Derive the solubility product expression for slightly soluble AgBr, CaCO3, Al(OH)3 and PbCrO4.

The solubility product (Ksp) is the product of the ion concentrations in a saturated solution of a sparingly soluble salt, each raised to its coefficient. AgBr ⇌ Ag+ + Br-: Ksp = [Ag+][Br-]. CaCO3 ⇌ Ca2+ + CO3^2-: Ksp = [Ca2+][CO3^2-]. Al(OH)3 ⇌ Al3+ + 3OH-: Ksp = [Al3+][OH-]^3. PbCrO4 ⇌ Pb2+ + CrO4^2-: Ksp = [Pb2+][CrO4^2-].

Q5. Explain the following. i) (a) State the Law of Mass Action and derive the equilibrium constant expression for the given equation. A + B ⇌ C + D (b) Given the equilibrium concentrations of N2, O2 and NO as 0.05 M, 0.05 M, and 5.5 x 10^-4 M respectively. Calculate Kc for the

decomposition of NO at 1500 degrees C for the reaction. ii) Find the value of equilibrium constant Kc for following reaction. 2SO2(g) + O2(g) ⇌ 2SO3(g). The equilibrium concentrations are [SO2] = 0.90 M, [O2] = 0.35 M and [SO3] = 1.1 M. (Ans: 4.3) iii) (a) State Le-Chatelier's principle and discuss its application to ammonia system. (b) What is the effect of raising temperature on each of the following equilibria. i) N2 + 3H2 ⇌ 2NH3, delta H = -ve; ii) 2NO + O2 ⇌ 2NO2, delta H = -ve; iii) 2SO2 + O2 ⇌ 2SO3, delta H = -ve; iv) N2 + O2 ⇌ 2NO, delta H = +ve. iv) (a) With the help of chemical equilibrium constant expression, how will you predict the direction and the extent of a chemical reaction? (b) The equilibrium constant for the reaction, 2O3 ⇌ 3O2 is 1.0 x 10^50 at 25 degrees C. Predict the extent of the formation of O2 at room temperature.

(i)(a) The Law of Mass Action states that the rate of a reaction is proportional to the product of the active masses (molar concentrations) of the reactants. For A + B ⇌ C + D, forward rate = kf[A][B] and reverse rate = kr[C][D]; at equilibrium the rates are equal, so kf[A][B] = kr[C][D], giving Kc = kf/kr = [C][D] / ([A][B]). (i)(b) For the decomposition 2NO ⇌ N2 + O2, Kc = [N2][O2] / [NO]^2 = (0.05 x 0.05) / (5.5 x 10^-4)^2 = 0.0025 / 3.025 x 10^-7 = 8.26 x 10^3 (the book's printed exponent reads 10^9 in the scan, which does not follow from the given concentrations; the worked value is 10^3). (ii) Kc = [SO3]^2 / ([SO2]^2 [O2]) = (1.1)^2 / ((0.90)^2 x 0.35) = 1.21 / 0.2835 = 4.27, about 4.3 as printed. (iii)(a) Le-Chatelier's principle states that if a stress is applied to a system at equilibrium, the equilibrium shifts to reduce the stress. For ammonia, N2 + 3H2 ⇌ 2NH3 is exothermic with 4 moles becoming 2, so low temperature and high pressure favour ammonia, and an iron catalyst is used to reach equilibrium faster; industry uses about 450 degrees C and 200 atm as a compromise. (iii)(b) Raising the temperature shifts an

exothermic forward reaction backwards and an endothermic forward reaction forwards: (i), (ii) and (iii) are exothermic in the forward direction, so heating reduces NH3, NO2 and SO3; (iv) is endothermic in the forward direction, so heating increases NO. (iv)(a) Compare the reaction quotient Q with Kc: if Q < Kc the reaction goes forward, if Q > Kc it goes backward, and if Q = Kc it is at equilibrium; a very large Kc means the reaction goes almost to completion, and a very small Kc means it hardly proceeds. (iv)(b) Kc = 1.0 x 10^50 is enormous, so the reaction goes virtually to completion: at room temperature ozone is almost completely converted into oxygen.

Project

Project. Take two test tubes, add about 10 cm^3 solution of potassium chromate (K2CrO4) in both of them. Note the colour of the solution. Place one test tube in test tube stand for colour comparison purpose and add dil. HCl solution to the second test tube till the colour changes from yellow to orange. Now add a few cm^3 solution of dil. NaOH. You will observe that the yellow colour appears again. Compare the colour with that of the first test tube solution. The reaction that is taking place due to the addition of acid is: 2CrO4^2-(aq) + 2H+(aq) ⇌ Cr2O7^2-(aq) + H2O (Yellow to Orange). Questions: i) Which principle is governing this reaction to bring it from left to right and back? ii) Why the reaction go back from right to left by the addition of base?

Step 1: Adding acid increases H+ ions, so by Le-Chatelier's principle the equilibrium 2CrO4^2- + 2H+ ⇌ Cr2O7^2- + H2O shifts to the right, and the yellow chromate turns into orange dichromate. Step 2: Adding base removes H+ ions (forming water), so the equilibrium shifts back to the left, and the yellow chromate colour returns. Step 3: Answers: (i)

Le-Chatelier's principle governs the shifts. (ii) The base consumes H+ ions, so the system replaces them by shifting left, back to chromate.

Unit 8 — Acids, Bases and Salts

Exercise — Choose the Correct Option

Q1. Which one is the example of buffer

a) HCl/NaCl

b) NaOH/H2CO3

c) NH4OH/NH4Cl [Correct]

d) NaOH/NaCl

Q2. Conjugate acid-base pair differs by

a) A proton [Correct]

b) A proton pair

c) An electron

d) An electron pair

Q3. 1 M solution of Ca(OH)2 is mixed with 1M solution of HCl. The product solution is

a) Acidic

b) Basic [Correct]

c) Neutral

d) Amphoteric

Q4. Cl- is the conjugate base of

a) AlCl3

b) NaCl

c) HCl [Correct]

d) KCl

Q5. pH of an aqueous solution is 9. Its pOH is

a) 11

b) 9

c) 7

d) 5 [Correct]

Q6. Salt of a weak base and strong acid has a pH, approximately

a) 8

b) 6 [Correct]

c) 7

d) 9

Q7. The unit of Kw is,

a) Mole .dm^-3

b) Mole^-2 .dm^-6

c) Mole^2 .dm^-6 [Correct]

d) Mole^2 .dm^-3

Q8. Very large Ka value means that the substance is a

a) Strong acid [Correct]

b) Weak acid

c) Weak base

d) Strong base

Q9. In following halogen acids which one is the strongest acid

a) HF

b) HCl

c) HBr

d) HI [Correct]

Q10. Which one of following solution have zero pH

a) 1M HCl [Correct]

b) 0.5M H2SO4

c) 0.1M HNO3

d) 1M CH3COOH

Q11. Which one is not true for acids

a) Liberate H+

b) Accepts electrons

c) Have high pH [Correct]

d) Turn blue litmus red

Q12. 10^-3 moles of HNO3 is dissolved /dm^3. Its pH is

a) -3

b) 5

c) 3 [Correct]

d) 1

Q13. A solution with a pKa value of 9, suggest that it is a

a) Strong acid

b) Weak acid [Correct]

c) Weak base

d) Strong base

Q14. Cl- is a / an

a) Acid

b) base [Correct]

c) Amphoteric

d) None

Q15. An acidic buffer solution can be prepared by mixing

a) Weak acid and its salt with strong base [Correct]

b) Strong acid and its salt with weak base

c) Weak base and its salt with strong acid

d) Strong base and its salt with weak acid

Exercise — Short Questions

Q1. What information would you use to support the view that water can

act either as a weak acid or as a weak base?

Water is amphoteric: it donates a proton to a strong base, acting as a weak acid ($H_2O + NH_3 \rightleftharpoons NH_4^+ + OH^-$), and it accepts a proton from a strong acid, acting as a weak base ($H_2O + HCl \rightleftharpoons H_3O^+ + Cl^-$). Its self-ionisation, $2H_2O \rightleftharpoons H_3O^+ + OH^-$, shows both roles at once, and its Kw of 1×10^{-14} shows both tendencies are very small.

Q2. Explain that why the sum of pKa and pKb is always equal to 14.

For a conjugate acid-base pair, $K_a \times K_b = K_w = 1 \times 10^{-14}$ at 25 degrees C. Taking negative logarithms gives $pK_a + pK_b = pK_w = 14$. The sum is always 14 because the two constants describe the same proton-transfer equilibrium from opposite sides, and their product is fixed by the ionisation of water.

Q3. Explain why the conjugate base of a strong acid is a weak base and the conjugate acid of a strong base is a weak acid.

A strong acid ionises completely, so its conjugate base has almost no tendency to accept a proton again, making it a very weak base; for example Cl^- from HCl. A strong base likewise takes a proton completely, so its conjugate acid has almost no tendency to lose a proton again, making it a very weak acid; for example Na^+ and water from NaOH. Strength on one side of a conjugate pair always means weakness on the other.

Q4. Justify your answer with equations that CH_3COONa gives a basic solution while NH_4Cl an acidic solution in water.

Sodium acetate hydrolyses: $CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$, producing hydroxide ions, so the solution is basic. Ammonium chloride hydrolyses: $NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$, producing hydrogen ions, so the solution is acidic. The anion of a weak acid gives a basic solution and the cation of a weak base gives an acidic solution.

Q5. Why do you call AlCl3 and BF3 as Lewis acids, Cl- and NH3 as Lewis bases?

AlCl3 and BF3 have incomplete octets and vacant orbitals that can accept an electron pair, so they are Lewis acids. Cl- and NH3 have lone pairs of electrons that they can donate to form a coordinate bond, so they are Lewis bases. The classification follows the electron-pair definition, not proton transfer.

Exercise — Numerical Questions

Q1. What is the pH of 0.0001M Ca(OH)2 solution. (Ans. pH = 10.3)

Step 1: [OH-] = 2 x 0.0001 = 2 x 10^-4 mol/dm^3. Step 2: pOH = -log(2 x 10^-4) = 3.70. Step 3: pH = 14 - 3.70 = 10.30. Answer: pH = 10.3, as printed in the book.

Q2. What is [H+] and [OH-] ions concentration of solution, which has a pH of 4.87? (Ans. [H+] = 1.35 x 10^-5, [OH-] = 7.41 x 10^-10)

Step 1: [H+] = 10^-pH = 10^-4.87 = 1.35 x 10^-5 mol/dm^3. Step 2: [OH-] = Kw/[H+] = 1 x 10^-14 / 1.35 x 10^-5 = 7.41 x 10^-10 mol/dm^3. Answers: [H+] = 1.35 x 10^-5, [OH-] = 7.41 x 10^-10 (the book's printed exponent for [OH-] reads 10^-8 in the scan, which does not follow from its own [H+] value; the worked value is 10^-10).

Q3. What is the pH of a 1.0 x 10^-4 M KOH solution? (Ans. pH = 10)

Step 1: [OH-] = 1.0 x 10^-4 mol/dm^3. Step 2: pOH = -log(1.0 x 10^-4) = 4. Step 3: pH = 14 - 4 = 10. Answer: pH = 10, as printed in the book.

Q4. What is the pH of a solution if the [H3O+] is 6.7 x 10^-4 M. (Ans. pH = 3.17)

Step 1: pH = -log[H3O+] = -log(6.7 x 10^-4). Step 2: pH = 3.17. Answer: pH = 3.17, as printed in the book.

Q5. What is the pH of a solution for which [OH-] is 0.15M. (Ans. pH =

13.2)

Step 1: pOH = -log[OH-] = -log(0.15) = 0.82. Step 2: pH = 14 - 0.82 = 13.18. Answer: pH = 13.2, as printed in the book.

Exercise — Descriptive Questions

Q1(a). What is Bronsted-Lowry acids and bases? Explain it with suitable examples.

A Bronsted-Lowry acid is a proton donor and a Bronsted-Lowry base is a proton acceptor. For example, in HCl + NH3 ⇌ NH4+ + Cl-, HCl donates a proton so it is the acid, and NH3 accepts it so it is the base. In CH3COOH + H2O ⇌ CH3COO- + H3O+, acetic acid is the acid and water is the base.

Q1(b). Write equations and indicate the conjugate acid-base pairs for the following; (i) Acetic acid and water, (ii) Ammonia and hydrochloric acid

(i) Acetic acid and water: CH3COOH + H2O ⇌ CH3COO- + H3O+. Conjugate pairs: CH3COOH/CH3COO- and H3O+/H2O. (ii) Ammonia and hydrochloric acid: NH3 + HCl ⇌ NH4+ + Cl-. Conjugate pairs: NH4+/NH3 and HCl/Cl-. In each pair the two members differ by one proton.

Q1(c). Justify that NH3 is a base according to Lewis concept.

Ammonia has a lone pair of electrons on nitrogen, which it can donate to an electron-pair acceptor to form a coordinate bond, for example NH3 + H+ → NH4+ and NH3 + BF3 → H3N-BF3. Donating an electron pair is the Lewis definition of a base, so NH3 is a Lewis base.

Q2(a). Define buffer solution. What is buffer action and show with equations how a buffer system works?

A buffer solution resists changes in pH when small amounts of strong acid or base are added. Buffer action works because the buffer contains a weak acid and its conjugate base (or a weak base and its conjugate acid) in equilibrium. In an acetic acid/acetate buffer, added H+ is removed by

CH3COO- + H+ → CH3COOH, and added OH- is removed by CH3COOH + OH- → CH3COO- + H2O, so the pH stays almost constant.

Q2(b). What are the applications of buffers solutions?

Buffer solutions are used to keep the pH of blood constant at about 7.4, in biochemical and pharmaceutical preparations, in electroplating baths, in photography, in leather tanning, and in agriculture to control soil pH. Any process that needs a steady pH uses a buffer.

Q2(c). Justify that buffer solution resists changes in pH, when a small amount of an acid or a base is added.

A buffer resists pH change because it contains both members of a conjugate pair. Added acid is consumed by the basic member and added base by the acidic member, converting the strong acid or base into a weak acid or base that barely changes the hydrogen ion concentration. Only when one member is used up does the pH change sharply.

Q3(a). Briefly describe the leveling effect.

The levelling effect is that in water, all acids stronger than H3O+ appear equally strong, and all bases stronger than OH- appear equally strong, because water converts them fully into H3O+ or OH-. For example, HCl, HBr and HClO4 all ionise completely in water, so their relative strengths cannot be distinguished in aqueous solution; a solvent like acetic acid is needed to tell them apart.

Q3(b). What is the relationship between Ka and Kb?

For a conjugate acid-base pair, Ka x Kb = Kw. The acid constant measures proton loss and the base constant measures proton gain by the conjugate base, and multiplying the two equilibrium expressions cancels to give the ionisation of water, Kw = 1 x 10^-14 at 25 degrees C.

Q3(c). Write the equation relating Ka for a weak acid and Kb for its conjugate base. Use NH3 and its conjugate acid NH4+ to derive the

relationship between Ka and Kb.

For NH3 and its conjugate acid NH4+: NH4+ ⇌ NH3 + H+, Ka = [NH3][H+]/[NH4+]. For the base: NH3 + H2O ⇌ NH4+ + OH-, Kb = [NH4+][OH-]/[NH3]. Multiplying gives Ka x Kb = [H+][OH-] = Kw = 1 x 10^-14. Taking negative logs gives pKa + pKb = 14.

Q4(a). What is meant by the term amphoteric? Give an example of a substance or ion with amphoteric characteristics.

Amphoteric means able to act as either an acid or a base. Water is amphoteric: it donates a proton to NH3 and accepts a proton from HCl. Other examples are the hydrogen carbonate ion HCO3-, which can lose a proton to give CO3^2- or gain one to give H2CO3, and zinc oxide ZnO, which reacts with both acids and bases.

Q4(b). Define pH, pOH, pKa and pKb.

pH = -log[H+], the negative logarithm of the hydrogen ion concentration. pOH = -log[OH-], the negative logarithm of the hydroxide ion concentration. pKa = -log Ka, measuring acid strength: a smaller pKa means a stronger acid. pKb = -log Kb, measuring base strength: a smaller pKb means a stronger base.

Q4(c). Explain ionization constant of water and calculate pH and pOH in aqueous medium using given Kw values.

Water ionises as 2H2O ⇌ H3O+ + OH-, with Kw = [H3O+][OH-] = 1 x 10^-14 at 25 degrees C. In pure water [H+] = [OH-] = sqrt(Kw) = 1 x 10^-7 mol/dm^3, so pH = 7 and pOH = 7. In any aqueous medium, pH + pOH = 14, so knowing Kw lets us calculate one from the other.

Q5(a). Define salt hydrolysis. Categorize salts according to how they affect the pH of a solution.

Salt hydrolysis is the reaction of the ions of a salt with water to give an acidic or basic solution. Salts fall into four groups: strong acid + strong

base, like NaCl, giving a neutral solution (pH 7); weak acid + strong base, like CH3COONa, giving a basic solution; strong acid + weak base, like NH4Cl, giving an acidic solution; and weak acid + weak base, like CH3COONH4, whose pH depends on whether Ka or Kb is larger.

Q5(b). (i) What are conjugate acids and bases? Give the conjugate bases of the following acids; HClO4, HCN, H2CO3, NH4+ (ii) Classify as acids and bases giving reasons; BF3, NH3, NH4+, Ag+, CaO, KCN, H2S, SO4^2-, Na+, Cl- (iii) Classify the following as Lewis acid or Lewis base; CO2, H2O, SO2, I-, NH3, OH-, BCl3

Conjugate acids and bases differ by one proton. The conjugate bases are: HClO4 → ClO4-, HCN → CN-, H2CO3 → HCO3-, NH4+ → NH3. Classified as acids and bases: BF3 is a Lewis acid (electron-pair acceptor); NH3 is a Lewis base (electron-pair donor); NH4+ is a Bronsted acid (proton donor); Ag+ is a Lewis acid; CaO is a base (oxide ion acceptor of protons); KCN gives CN-, a base; H2S is an acid; SO4^2- is a base; Na+ is neutral; Cl- is a weak base. As Lewis acids or bases: CO2 is a Lewis acid, H2O a Lewis base, SO2 a Lewis acid, I- a Lewis base, NH3 a Lewis base, OH- a Lewis base, BCl3 a Lewis acid.

Project

Project i. Arrange the following common substances in order of increasing pH: Eggs, Apple, Tomato, Milk, Banana, Potatoes, Lemon, Shampoo, Water, Carbonated drink.

Step 1: Test or look up approximate pH values: lemon about 2, carbonated drink about 3, apple about 3.5, tomato about 4, banana about 5, potatoes about 6, milk about 6.5, water 7, shampoo about 8, eggs about 8. Step 2: Arrange in increasing pH: lemon, carbonated drink, apple, tomato, banana, potatoes, milk, water, shampoo, eggs. Step 3: Note that exact

values vary with variety and brand, but the order from most acidic to most basic stays the same.

Project ii. Group Work and Discussion: Make a buffer solution in the laboratory. Record the pH of the buffer solution. Add small amount of strong acid to the buffer solution and record the pH of the solution. Add small amount of the strong base to the buffer solution and record the pH of the solution. Explain how such solution maintains a constant pH, even with the addition of small amounts of strong acid or strong base. Present your group work in the class and answer the questions of your classmates.

Step 1: Prepare a buffer in the laboratory, for example acetic acid with sodium acetate, and record its pH. Step 2: Add a small amount of strong acid and record the pH again; it changes very little. Step 3: Add a small amount of strong base to a fresh sample and record the pH; again it changes very little. Step 4: Explain that the buffer's conjugate pair consumes the added H+ or OH-, keeping the pH nearly constant. Step 5: Present the group work in class and answer questions.

Unit 9 — Chemical Kinetics

Exercise — Choose the Suitable Option

Q(i). Activated complex is a substance which is

a) stable

b) unstable [Correct]

c) can be isolated

d) can exist as product

Q(ii). A reaction is first order with respect to A and second order with respect to B, the rate equation is

a) rate = k [A][B]^2 [Correct]

b) rate = k [A]^2[B]

c) rate = k [A][B]

d) rate = k [A]

Q(iii). For a reaction A -> Product, doubling the concentration of A, quadruples the rate. The reaction is;

a) first order

b) second order [Correct]

c) zero order

d) third order

Q(iv). With the increase in temperature the rate of reaction increases due to

a) decrease in collisions

b) decrease in activation energy

c) increase in kinetic energy of the molecules

d) more number of molecules attains activation energy. [Correct]

Q(v). You can speed up a reaction by:

a) using less concentrated solution.

b) by lowering the temperature.

c) using more concentrated solution. [Correct]

d) stopping stirring.

Q(vi). When the catalyst and reactants are in different physical phases, we call the catalysis:

a) Heterogeneous [Correct]

b) Homogeneous

c) Concentrated

d) positive catalysis

Q(vii). Increasing the temperature increases the rate of reaction by

a) lowering the activation energy.

b) increasing the activation energy.

c) lowering the frequency of effective collisions between reacting molecules.

d) increasing the frequency of effective collisions between reacting molecules. [Correct]

Q(viii). The energy needed to start the reaction is called

a) potential energy

b) activation energy [Correct]

c) kinetic energy

d) ionization energy

Q(ix). As the number of effective collisions between the reacting particles increases, the rate of reaction

a) Decreases

b) Increases [Correct]

c) remains the same

d) both b and c

Q(x). Which condition will increase the rate of reaction?

a) Decrease in temperature and concentration of reactants.

b) Decrease in temperature and increase in concentration of reactants.

c) Increase in temperature and decrease in concentration of reactants.

d) Increase in temperature and increase in concentration of reactants. [Correct]

Q(xi). A catalyst

a) decrease the activation energy and decrease the rate of reaction.

b) decrease the activation energy and increase the rate of reaction. [Correct]

c) increase the activation energy and decrease the rate of reaction.

d) increase the activation energy and increase the rate of reaction.

Q(xii). Which aluminum sample would react most rapidly with 1 M CuSO4 solution under the same conditions?

a) 1 g Al rod

b) 1 g Al pellets

c) 1 g Al ribbon

d) 1 g Al powder [Correct]

Q(xiii). One mole of a reactant reacts with a rate of 0.6 mol dm^-3 s^-1. What is the rate constant of this reaction?

a) 1 s^-1

b) 0.3 s^-1

c) 0.6 s^-1 [Correct]

d) 0.9 s^-1

Exercise — Short Questions

Q(i). Determine the overall orders from the following rate equations (a) rate = k[NO]^2[O2] (b) rate = k[NO]^2

Overall order is the sum of the powers in the rate equation. (a) rate = k[NO]^2[O2]: order = 2 + 1 = 3, third order. (b) rate = k[NO]^2: order = 2, second order.

Q(ii). Explain why a molecular collision should be sufficiently energetic to cause a reaction.

A collision causes a reaction only if the molecules collide with enough kinetic energy to break old bonds and form the activated complex, that is, energy equal to or greater than the activation energy. Slow molecules simply bounce apart unchanged, however often they collide.

Q(iii). Name the four factors that increase the rate of reaction.

The four factors are the nature of the reactants, the concentration of the reactants, the temperature, and the presence of a catalyst. For solid

reactants, the particle size (surface area) is a fifth factor.

Q(iv). In the light of collision theory of reaction rate, give reasons of increase in the rate by the increase of temperature.

Collision theory says reaction occurs when molecules collide with sufficient energy and proper orientation. Raising the temperature increases molecular speeds, so collisions are more frequent, and more importantly a much larger fraction of molecules now has energy above the activation energy, so the number of effective collisions rises sharply and the rate increases.

Q(v). Consider two gases, A and B, in a container at room temperature. What effect would the following changes have on the rate of the reaction between these gases? (a) The pressure is doubled. (b) The number of molecules of gas A is doubled. (c) The temperature is decreased by 10 degrees C

(a) Doubling the pressure doubles the concentrations of both gases, so collisions become more frequent and the rate increases; for a reaction first order in each gas the rate would increase four times. (b) Doubling the number of molecules of gas A doubles its concentration, so the rate increases in proportion to the order in A. (c) Decreasing the temperature by 10 degrees C lowers the fraction of molecules with activation energy, so the rate falls, roughly to half for many reactions.

Q(vi). The rate constant for the reaction CO + NO2 -> CO2 + NO at 400 degrees C is 0.50 dm^3 mol^-1 sec^-1, and the reaction is first order with respect to both CO and NO2. (a) What is the overall order of the reaction? (b) What is the rate of the reaction at 400 degrees C when the concentration of CO is 0.025 mol dm^-3 and that of NO2 is 0.040 mol dm^-3? (Ans: 5 x 10^-4 mol dm^-3 sec^-1)

(a) Overall order = 1 + 1 = 2, second order. (b) Step 1: rate = k[CO][NO2]

= 0.50 x 0.025 x 0.040. Step 2: rate = 5 x 10^-4 mol dm^-3 s^-1.

Answer: 5 x 10^-4 mol dm^-3 s^-1, as printed in the book.

Q(vii). Explain briefly why all collisions between reactant molecules do not lead to reaction?

Most collisions are not effective because the molecules may collide with too little energy to reach the activation energy, or they may collide in the wrong orientation, so the bonds that must break are not struck. Only collisions that are both energetic enough and correctly oriented form the activated complex and lead to products.

Exercise — Descriptive Questions

Q1. Define the following terms. (a) Rate of the reaction (b) Rate constant. (c) Order of the reaction. (d) Rate law or rate equation. (e) Catalyst

Rate of the reaction is the change in concentration of a reactant or product per unit time, in mol dm^-3 s^-1. Rate constant (k) is the proportionality constant in the rate equation; it equals the rate when all reactant concentrations are 1 mol/dm^3, and it depends on temperature. Order of the reaction is the sum of the powers of the concentration terms in the rate equation, found by experiment. Rate law (rate equation) is the equation relating the rate to the molar concentrations of reactants raised to experimentally determined powers, for example rate = k[A][B]. A catalyst is a substance that changes the rate of a reaction without itself being consumed.

Q2(a). Define the activation energy. What role does it play in chemical reactions?

Activation energy is the minimum extra energy that reactant molecules must gain to form the activated complex and react. It acts as an energy barrier: only molecules with energy equal to or above it can react, so the

size of the barrier decides how fast a reaction goes. A catalyst works by providing a path with a lower activation energy.

Q2(b). Household gas (methane) burns in the presence of oxygen and gives energy for our daily use. If a mixture of methane and oxygen is kept in a container for indefinite period, no reaction takes place, explain?

Methane and oxygen do not react at room temperature because their molecules do not have enough energy to cross the activation energy barrier, even though the reaction would release energy. A spark or flame supplies the activation energy to a few molecules, and the heat released then activates more molecules, so the burning continues by itself.

Q3. Explain how does a catalyst increase the rate of a reaction? Compare catalyzed and un-catalyzed reaction on a potential energy diagram.

A catalyst increases the rate by providing an alternative reaction path with a lower activation energy, so more molecules can cross the barrier at the same temperature. On a potential energy diagram the catalysed path rises to a lower peak (lower activated complex) than the uncatalysed path, while the energies of reactants and products, and therefore delta H, stay the same. The catalyst is regenerated at the end and is not consumed.

Q4. Explain the relationship between reactant concentration and rate of reaction.

The rate of a reaction generally increases when the concentration of a reactant increases, because more reactant particles mean more collisions per second. The exact relation is given by the rate law: for a first-order reaction doubling the concentration doubles the rate, and for a second-order reaction doubling the concentration quadruples the rate. If the order in a reactant is zero, changing its concentration does not change the rate.

Q5. Discuss why, (according to collision theory of chemical reactions) some molecular collisions result in a chemical reaction while others do not. Draw and explain a reaction energy diagram (reaction profile) for (a) an exothermic reaction. (b) an endothermic reaction. Also show in the diagram (a) position of the energy of reactants (b) position of the energy of products (c) activation energy. (d) delta H of the reaction.

According to collision theory, a reaction happens only when molecules collide with energy at least equal to the activation energy and with the correct orientation. Most collisions fail one of these tests, so they do not react. The reaction energy diagram (reaction profile) plots potential energy against the progress of the reaction: for an exothermic reaction the products lie lower than the reactants, delta H is negative, and the peak between them is the activated complex, whose height above the reactants is the activation energy. For an endothermic reaction the products lie higher than the reactants, delta H is positive, and the peak is again the activated complex. In both diagrams the positions to mark are: (a) energy of reactants, (b) energy of products, (c) activation energy, (d) delta H of the reaction.

Exercise — Numerical

Q1. The conversion of X to Y follows second order kinetics. If concentration X is increased 3 times to what factor the rate of formation of Y will be increased. (Ans. 9 times)

Step 1: For second-order kinetics, rate is proportional to [X]^2. Step 2: Increasing [X] by 3 times multiplies the rate by 3^2 = 9. Answer: 9 times, as printed in the book.

Q2. A first order reaction (A -> Product) has a rate of 0.008 Ms^-1 with [A] = 0.002 M, what is the rate constant for this reaction? (Ans. 4 s^-1)

Step 1: For a first-order reaction, rate = k[A]. Step 2: k = rate/[A] = 0.008 / 0.002 = 4 s^-1. Answer: 4 s^-1, as printed in the book.

Q3. (a) The decomposition of N2O5 proceeds according to the following equation. 2N2O5(g) -> 4NO2(g) + O2(g) (b) If the rate of decomposition of N2O5 at a particular instant is 4.2 x 10^-7 Ms^-1, what is the rate of appearance of (a) NO2 (b) O2? (Ans. (a) 8.4 x 10^-7 Ms^-1 (b) 2.1 x 10^-7 Ms^-1)

Step 1: From 2N2O5(g) -> 4NO2(g) + O2(g), the rate is (1/2) x rate of disappearance of N2O5 = (1/4) x rate of appearance of NO2 = rate of appearance of O2. Step 2: Rate of appearance of NO2 = 2 x 4.2 x 10^-7 = 8.4 x 10^-7 Ms^-1. Step 3: Rate of appearance of O2 = (1/2) x 4.2 x 10^-7 = 2.1 x 10^-7 Ms^-1. Answers as printed in the book.

Project

Project. Hydrogen peroxide decomposes to produce oxygen and water according to reaction shown below. 2H2O2(aq) -> 2H2O(l) + O2(g). The reaction is quite slow unless catalysed by substances such as iodide ions, manganese dioxide, ferric ions etc. Take a test tube and add about 10 to 15 cm^3 of hydrogen peroxide in it. Add a few crystals of potassium iodide to it and watch the bubbles of oxygen coming out from the test tube. Bring a glowing match stick at the mouth of the test tube and observe. What do you expect to happen with the stick? Explain.

Step 1: Take a test tube with 10 to 15 cm^3 of hydrogen peroxide and add a few crystals of potassium iodide. Step 2: Oxygen bubbles come out rapidly because the iodide catalyses the decomposition 2H2O2(aq) -> 2H2O(l) + O2(g). Step 3: Bring a glowing match stick to the mouth of the test tube: it bursts back into flame, because oxygen supports combustion. Step 4: Explain that without the catalyst the same reaction is very slow,

and that the catalyst is not consumed.

Unit 10 — Solutions and Colloids

Exercise — Choose the Correct Answer

Q1. In atmospheric gaseous solution, nitrogen is a

a) solvent [Correct]

b) solute

c) mixture

d) dispersed phase

Q2. In ice cream, sugar is

a) solute [Correct]

b) mixture

c) solvent

d) none

Q3. Elevation of boiling point is a property

a) additive

b) constitutive

c) colligative [Correct]

d) none

Q4. Change in enthalpy, delta H(sol), is heat of solution which is

a) positive

b) negative

c) zero

d) may be positive or negative. [Correct]

Q5. Freezing point of solution as compared to the solvent is

a) higher

b) lower [Correct]

c) variable

d) remains the same

Q6. Phenol - water system is the example of

a) completely miscible liquids

b) completely immiscible liquids

c) partially miscible liquids [Correct]

d) none

Q7. Fog is an example of

a) solution

b) colloid [Correct]

c) suspension

d) coarse mixture

Q8. Which of the following solute would not form a solution if added to water?

a) KCl

b) CH4 [Correct]

c) HCl

d) NH3

Q9. Vapour pressure of a given liquid will decrease if

a) surface area of liquid is increased.

b) total volume of liquid in the container is decreased.

c) volume of the vapour phase is increased.

d) the temperature is decreased. [Correct]

Q10. For a dilute solution Raoult's law states that

a) The lowering of vapour pressure is equal to the mole fraction of solute.

b) The relative lowering of vapour pressure is equal to the mole fraction of solute. [Correct]

c) The relative lowering of vapour pressure is proportional to the amount of solute.

d) The vapour pressure of the solution is equal to the mole fraction of solvent.

Q11. The boiling points of C6H6, CH3OH, C6H5NH2, and C6H5NO2 are 80 degrees C, 65 degrees C, 184 degrees C and 112 degrees C respectively. Which one will show the highest vapour pressure at room temperature?

a) C6H6

b) CH3OH [Correct]

c) C6H5NH2

d) C6H5NO2

Q12. The process in which the water molecules surround and interact with the solute ions or molecules is called

a) neutralization

b) hydration [Correct]

c) hydrolysis

d) hydrogenation

Q13. The molarity of pure water in itself is

a) 18

b) 50

c) 59.5

d) 55.6 [Correct]

Exercise — Short Questions

Q1. Define the following concentration units: i) ppm ii) ppb iii) ppt iv) mole fraction

ppm (parts per million) is the number of parts of solute per million parts of solution, by mass or volume. ppb is parts per billion, and ppt is parts per trillion, used for very dilute solutions. Mole fraction of a component is its moles divided by the total moles of all components in the solution, and

it has no units.

Q2. Which concentration unit, from molarity and molality, depends on temperature.

Molarity depends on temperature because it is moles per volume of solution, and the volume of a solution expands or contracts with temperature. Molality is moles per kilogram of solvent, and since mass does not change with temperature, molality is independent of temperature.

Q3. Prove mathematically that sum of the mole fractions of all the components of a solution is equal to one.

For a solution of two components A and B, mole fraction of A is XA = nA/(nA + nB) and of B is XB = nB/(nA + nB). Adding them gives XA + XB = (nA + nB)/(nA + nB) = 1. The same holds for any number of components: the sum of all mole fractions is always one.

Q4. Define and name four colligative properties.

Colligative properties depend only on the number of solute particles, not on their nature. The four are: lowering of vapour pressure, elevation of boiling point, depression of freezing point, and osmotic pressure.

Q5. Define osmosis and give it applications in daily life.

Osmosis is the flow of solvent molecules through a semi-permeable membrane from a dilute solution to a concentrated solution. In daily life it explains why potato chips swell in fresh water and shrink in salt solution, why plant roots absorb water from soil, why red blood cells burst in distilled water and shrink in salt solution, and it is used in reverse osmosis plants to desalinate sea water.

Exercise — Numerical

Q1. Calculate Molarity (M) of the following solutions: i) 2.0 g of H2SO4/2

dm^3 of aqueous solution ii) 0.4 g of NaOH/100 cm^3 of aqueous solution iii) 0.5 g of Na2CO3/250 cm^3 of aqueous solution. Ans. (0.01M, 0.1M, 0.02M)

Step 1: (i) Moles of H2SO4 = 2.0/98 = 0.0204 mol, volume = 2 dm^3, so M = 0.0204/2 = 0.01 M. Step 2: (ii) Moles of NaOH = 0.4/40 = 0.01 mol, volume = 100 cm^3 = 0.1 dm^3, so M = 0.01/0.1 = 0.1 M. Step 3: (iii) Moles of Na2CO3 = 0.5/106 = 0.00472 mol, volume = 250 cm^3 = 0.25 dm^3, so M = 0.00472/0.25 = 0.02 M (rounded). Answers as printed in the book: (0.01M, 0.1M, 0.02M).

Q2. Calculate mole fraction of each component in the following solutions. i) 2.5 moles of water (H2O) + 1.5 moles of NaCl. ii) 500 g of water (H2O) + 600 g of glucose (C6H12O6). iii) 800 g of water (H2O) + 120 g of acetone (CH3)2CO. Ans. (0.625, 0.375), (0.89, 0.11), (0.955, 0.044)

Step 1: (i) X(water) = 2.5/(2.5 + 1.5) = 0.625, X(NaCl) = 1.5/4.0 = 0.375. Step 2: (ii) Moles of water = 500/18 = 27.78, moles of glucose = 600/180 = 3.33, total = 31.11, so X(water) = 0.89, X(glucose) = 0.11. Step 3: (iii) Moles of water = 800/18 = 44.44, moles of acetone = 120/58 = 2.07, total = 46.51, so X(water) = 0.955, X(acetone) = 0.044. Answers as printed in the book.

Q3. A 3.0 cm^3 cylinder contains a mixture of oxygen, nitrogen, and chlorine gases. If their amounts are 4.64 x 10^-5 g, 3.5 x 10^-4 g and 2.3 x 10^-5 g respectively. Calculate the concentration of each in i) Parts per million (ppm) ii) Parts per billion (ppb). Answers: i) ppm O2 15.46, N2 116.6, Cl2 7.7; ii) ppb O2 1.546 x 10^4, N2 1.16 x 10^5, Cl2 7.7 x 10^3

Step 1: (i) ppm: O2 = (4.64 x 10^-5 / 3.0) x 10^6 = 15.46, N2 = (3.5 x 10^-4 / 3.0) x 10^6 = 116.6, Cl2 = (2.3 x 10^-5 / 3.0) x 10^6 = 7.7. Step 2: (ii) ppb = ppm x 1000: O2 = 1.546 x 10^4, N2 = 1.16 x 10^5, Cl2 = 7.7 x 10^3. Answers as printed in the book.

Q4. In an experiment 6.4 g of an organic compound was dissolved in 25.6 g of benzene and 1.0 degrees C depression of freezing point was observed. Calculate the molecular weight of the substance. (kf of benzene = 5.12) Ans. 1280 g/mol

Step 1: Molality m = (mass of solute / molar mass) / (mass of solvent in kg) = (6.4/M) / 0.0256 = 250/M. Step 2: delta Tf = Kf x m, so 1.0 = 5.12 x 250/M. Step 3: M = 1280 g/mol. Answer: 1280 g/mol, as printed in the book.

Q5. Calculate molecular weight of Iodine (I2) when a solution containing 1.19 g of I2 in 35.0 grams of ether shows a rise in boiling point by 0.296 degrees C (kb for ether: 2.22) (Ans. M.Wt of I2 = 254 g/mol)

Step 1: Molality m = (1.19/M) / 0.035 = 34/M. Step 2: delta Tb = Kb x m, so 0.296 = 2.22 x 34/M. Step 3: M = 255 g/mol, about 254 g/mol as printed in the book (the known molar mass of I2). Answer: M.Wt of I2 = 254 g/mol.

Exercise — Long Questions

Q1. How a given mixture can be differentiated into a true solution or coarse suspension.

A true solution has solute particles smaller than 1 nm, invisible even under a microscope, which do not settle, pass through filter paper, and do not show the Tyndall effect. A coarse suspension has particles larger than 1000 nm, visible to the eye, which settle on standing, are caught by filter paper, and scatter light. A colloid lies between the two, with particles of 1 to 1000 nm that show the Tyndall effect but do not settle.

Q2. Define solubility, what are the factors that affect the solubility of substances?

Solubility is the maximum amount of a solute that dissolves in a given

amount of solvent at a given temperature to form a saturated solution. It depends on the nature of the solute and solvent (like dissolves like), on temperature (most solids dissolve more in hot solvent, gases dissolve less), and on pressure for gases (Henry's law: higher pressure dissolves more gas).

Q3. Classify colloids on the basis of their behaviour towards medium and the physical states of matter.

By behaviour towards the medium, colloids are lyophilic (solvent-loving, stable and reversible, like starch in water) or lyophobic (solvent-hating, unstable and irreversible, like gold sol). By the physical states of the dispersed phase and medium, they are sols (solid in liquid), gels (liquid in solid), emulsions (liquid in liquid, like milk), foams (gas in liquid), and aerosols (liquid or solid in gas, like fog and smoke).

Q4. Give the statement of Raoult's law. Explain the lowering of vapour pressure of a solution based on this law.

Raoult's law states that the vapour pressure of a solution equals the mole fraction of the solvent times the vapour pressure of the pure solvent, P = X(solvent) x P0. When a non-volatile solute is added, some solvent molecules at the surface are replaced by solute particles, so fewer solvent molecules escape into the vapour, and the vapour pressure of the solution is lower than that of the pure solvent. The relative lowering equals the mole fraction of the solute.

Q5. Explain the phenomenon of osmosis.

Osmosis is the passage of solvent molecules through a semi-permeable membrane from a dilute solution into a concentrated solution. The membrane allows solvent molecules to pass but blocks solute particles, so solvent flows until the concentrations equalise or the osmotic pressure balances the flow. A thistle funnel with a membrane and sugar solution

rising in its stem demonstrates osmosis.

Q6. Explain the phenomenon of depression of freezing point. How molecular mass of solute can be calculated on the basis of depression in freezing point?

When a non-volatile solute is added to a solvent, the freezing point of the solution is lower than that of the pure solvent, because the solute particles hinder the formation of the solid crystal lattice. The depression is delta Tf = Kf x m, where Kf is the molal freezing-point constant and m is the molality. Since m = (1000 x w)/(M x W), the molar mass is M = (Kf x 1000 x w)/(delta Tf x W), where w is the mass of solute and W the mass of solvent in grams.

Q7. Explain the elevation of boiling point, how molecular mass of solute can be calculated on the basis of elevation of boiling point?

When a non-volatile solute is added to a solvent, the boiling point of the solution is higher than that of the pure solvent, because the vapour pressure is lowered and a higher temperature is needed for it to equal the atmospheric pressure. The elevation is delta Tb = Kb x m, so the molar mass is M = (Kb x 1000 x w)/(delta Tb x W), where w is the mass of solute and W the mass of solvent in grams.

Project

Project. Take a potato of four to five centimetres length. Peel off it and cut rod shaped finger chips of equal length. Measure the length of these chips. Place three chips in distilled water and three in concentrated sodium chloride solution. After one hour take out the chips and measure the length of the chips. Record the difference in length of the chips before and after dipping. You will observe that the chips in distilled water will be longer than before and those in the solution of NaCl will be shorter than

before. Guess what might be the reason for this observation?

Step 1: Cut equal finger-shaped chips from a potato, measure their lengths, and place three in distilled water and three in concentrated sodium chloride solution. Step 2: After one hour, take them out and measure again. Step 3: The chips in distilled water will be longer than before, because water enters the potato cells by endosmosis. Step 4: The chips in salt solution will be shorter than before, because water leaves the cells by exosmosis into the concentrated solution. Step 5: Conclude that osmosis moves water from a dilute to a concentrated solution across the cell membrane.

Unit 11 — Thermochemistry

Self-Assessment

Q1. Define calorimetry and how to measure enthalpy change with the help of calorimeter.

Calorimetry is the measurement of heat changes in chemical or physical processes using a calorimeter, an insulated vessel that prevents heat exchange with the surroundings. To measure an enthalpy change, the reaction is carried out in the calorimeter, the temperature change of the water (and calorimeter) is recorded, and the heat is calculated as q = m x c x delta T, where m is the mass, c the specific heat capacity and delta T the temperature change; at constant pressure this heat equals delta H.

Q2. State the Hess's law.

Hess's law of constant heat summation states that the total enthalpy change of a chemical reaction is the same whether the reaction takes place in one step or in several steps, because enthalpy is a state function.

Q3. On which principle Born Haber cycle is based?

The Born-Haber cycle is based on Hess's law: since enthalpy is a state

function, the enthalpy change of forming an ionic crystal can be found by adding the enthalpy changes of a cycle of steps (sublimation, ionisation, dissociation, electron affinity and lattice formation) that start and end at the same states.

Exercise — Choose the Correct Option

Q1. delta H per mole is expressed in the units of

a) kJ [Correct]

b) degrees F

c) degrees C

d) K

Q2. Which one of the following is NOT a state function?

a) enthalpy

b) heat [Correct]

c) temperature

d) pressure

Q3. For solids and liquids

a) delta H = delta E [Correct]

b) delta H > delta E

c) delta H < delta E

d) delta E = 0

Q4. 1 k cal is equal to

a) 41.8 x 10^3 J

b) 418 x 10^3 J

c) 4.18 x 10^3 J [Correct]

d) 0.418 x 10^3 J

Q5. Standard enthalpy (delta H0) for 1 mole of a substance, which exists in its natural state, at 1 atm. pressure is measured at

a) 0 K

b) 298 K [Correct]

c) 273 K

d) 0 degrees C

Q6. delta H can be measured indirectly by applying

a) Avogadro's law

b) Gas laws

c) Hess's law [Correct]

d) Faraday's law

Q7. Enthalpy means

a) Disorder

b) Transition state

c) Rate constant

d) Heat content [Correct]

Q8. No work is done at constant

a) Pressure

b) Volume [Correct]

c) Temperature

d) Mass

Q9. Heat capacity depends on

a) Pressure

b) Volume

c) Composition

d) Mass [Correct]

Q10. Which one of the following reactions is spontaneous?

a) Endothermic

b) Exothermic [Correct]

c) Reversible

d) Irreversible

Q11. The energy possessed by water in a dam is

a) Potential Energy [Correct]

b) Kinetic Energy

c) Electric Energy

d) Heat Energy

Q12. When heat is absorbed from the surrounding, the process is

a) Reversible

b) Mechanical

c) Exothermic

d) Endothermic [Correct]

Q13. The sum of all the energies of all the molecules or atoms of a substance is called its:

a) Specific heat

b) Heat capacity

c) Latent heat

d) Internal energy [Correct]

Q14. Which one of the following processes has delta H positive

a) Ionization energy [Correct]

b) Electron affinity

c) Combustion

d) Exothermic reaction

Exercise — Short Questions

Q1. Explain the following short questions with reasons. (i) Total energy of the system and its surroundings remains constant. (ii) Ionic reactions are very fast. (iii) The work done has positive and negative values. (iv) Heat of combustion is always negative. (v) Enthalpy change is a state function

but heat is not. (vi) delta H for solids and liquids becomes equal to delta E. (vii) Enthalpy of neutralization of strong acids and strong bases has always the same value.

(i) By the first law of thermodynamics, energy can neither be created nor destroyed, so any energy lost by the system is gained by the surroundings and the total energy of system plus surroundings stays constant. (ii) Ionic reactions are very fast because the ions are already free in solution and only need to come together by electrostatic attraction; no covalent bonds have to be broken first. (iii) Work done has a sign: work done by the system on the surroundings (expansion) is negative, and work done on the system by the surroundings (compression) is positive. (iv) Heat of combustion is always negative because burning always releases heat to the surroundings; combustion is always exothermic. (v) Enthalpy change depends only on the initial and final states, so it is a state function, while heat depends on the path taken (constant pressure or constant volume), so it is not. (vi) Since delta H = delta E + P delta V, and for solids and liquids the volume change delta V is almost zero, the P delta V term vanishes and delta H becomes equal to delta E. (vii) Neutralisation of any strong acid with any strong base is always the same reaction, $H^+ + OH^- \rightarrow H_2O$, so its enthalpy change is always the same, about -57.3 kJ/mol.

Q2. What do you mean by state of a system? What are state functions?

The state of a system is its condition described by measurable properties such as pressure, volume, temperature and composition. State functions are properties whose values depend only on the present state of the system, not on how it reached that state, for example pressure, volume, temperature, internal energy, enthalpy and entropy. Heat and work are not state functions because they depend on the path taken.

Exercise — Long Questions

Q1. State and explain Hess's law of constant heat summation. Show that it is a direct consequence of the first law of thermodynamics.

Hess's law states that the enthalpy change of a reaction is the same whether it occurs in one step or in a series of steps. It follows from the first law of thermodynamics because enthalpy is a state function: the change in a state function depends only on the initial and final states, not on the path. Since energy is conserved, the sum of the enthalpy changes of any set of steps joining the same reactants to the same products must equal the enthalpy change of the direct reaction.

Q2. How does Born Haber cycle help in calculating lattice energy?

The Born-Haber cycle is a closed cycle of steps that forms an ionic crystal from its elements: sublimation of the metal, ionisation of the metal atoms, dissociation of the non-metal molecules, electron affinity of the non-metal atoms, and finally lattice formation. By Hess's law the sum of all steps equals the enthalpy of formation of the crystal, so the unknown lattice energy can be calculated from the other known values. For example, using the book's worked values, lattice energy = -405.8 - 88 - 414 - (1/2)(192.5) - (-334.7) = -669.3 kJ/mol.

Q3. What is meant by heat of a reaction? Explain how the heat of a reaction at constant volume differs from the heat of a reaction at constant pressure?

The heat of a reaction is the heat absorbed or released when the reaction occurs. At constant volume no pressure-volume work is done, so the heat equals the change in internal energy, qv = delta E. At constant pressure the system may expand or contract, doing work, so the heat equals the change in enthalpy, qp = delta H. The two differ by the work term: delta H = delta E + P delta V = delta E + delta n RT for gases.

Q4. What is meant by pressure volume work? How it is related to the first law of thermodynamics?

Pressure-volume work is the work done when a system expands or contracts against an external pressure, w = -P delta V. It is related to the first law, delta E = q + w: the change in internal energy of a system equals the heat added to it plus the work done on it. When a gas expands, it does work on the surroundings, so w is negative and the internal energy falls unless heat is supplied.

Exercise — Numerical

Q1. Apply Hess's law to calculate delta H0 for the sublimation of one mole of iodine from the following equations. (i) H2(g) + I2(s) -> 2HI(g), delta H0 = 51.8 kJ/mol (ii) H2(g) + I2(g) -> 2HI(g), delta H0 = -10.5 kJ/mol (Ans. 62.3 kJ/mol)

Step 1: Sublimation is I2(s) -> I2(g). Subtract equation (ii) from equation (i): the H2 and HI cancel. Step 2: delta H = 51.8 - (-10.5) = 62.3 kJ/mol. Answer: 62.3 kJ/mol, as printed in the book.

Q2. Calculate heat of formation of an aqueous solution of NH4Cl from the following data. (i) NH3(g) + aq -> NH3(aq), delta H0 = -35.16 kJ/mol (ii) HCl(g) + aq -> HCl(aq), delta H0 = -72.41 kJ/mol (iii) NH3(aq) + HCl(aq) -> NH4Cl(aq), delta H0 = -51.48 kJ/mol (Ans. -159.08 kJ/mol)

Step 1: Add the three steps: NH3(g) + aq -> NH3(aq), HCl(g) + aq -> HCl(aq), NH3(aq) + HCl(aq) -> NH4Cl(aq). Step 2: The sum gives NH3(g) + HCl(g) + aq -> NH4Cl(aq). Step 3: delta H = -35.16 + (-72.41) + (-51.48) = -159.05 kJ/mol (the book prints -159.08 kJ/mol, the difference being rounding of the given values). Answer: about -159.1 kJ/mol.

Q3. Liquid ethanol when burnt in oxygen at 25 degrees C liberates heat i.e. delta H0 = -1402.14 kJ/mol. The heats of formation of CO2 and H2O

are -393.50 and -285.81 kJ/mol, respectively, at the same temperature. Calculate the heat of formation of ethanol at 25 degrees C. (Ans. -242.37 kJ/mol)

Step 1: Combustion: C2H5OH(l) + 3O2(g) -> 2CO2(g) + 3H2O(l), delta H = -1402.14 kJ/mol. Step 2: delta H = [2 x (-393.50) + 3 x (-285.81)] - delta Hf(ethanol). Step 3: -1402.14 = -1644.43 - delta Hf(ethanol). Step 4: delta Hf(ethanol) = -1644.43 + 1402.14 = -242.29 kJ/mol (the book prints -242.37 kJ/mol, the difference being rounding). Answer: about -242.3 kJ/mol.

Q4. A chemical reaction takes place in a container of cross sectional area of 0.01 m^2, fitted with a weightless and frictionless piston. The piston is moved up through 0.01 m against an external pressure of 101325 Pa as a result of the reaction. Calculate the work done by the system. (Ans. 101.3 J)

Step 1: Work w = P x delta V, and delta V = area x distance moved. Step 2: With the book's printed answer, delta V = 0.001 m^3, so w = 101325 x 0.001 = 101.3 J. NOTE: the displacement as read in the scan is 0.01 m with area 0.01 m^2, which gives delta V = 1 x 10^-4 m^3 and w = 10.13 J; the book's printed answer 101.3 J corresponds to a displacement of 0.1 m, so the printed distance and the printed answer differ by a factor of ten. Answer as printed in the book: 101.3 J.

Q5. Determine delta H0 for the following reaction with the help of given data. 2C(s) + 2H2(g) -> C2H4(g), delta H0 = ? i) C(s) + O2(g) -> CO2(g), delta H01 = -393.5 kJ/mol ii) H2(g) + (1/2)O2(g) -> H2O(g), delta H02 = -285.7 kJ/mol iii) C2H4(g) + 3O2(g) -> 2CO2(g) + 2H2O(g), delta H03 = -1430.9 kJ/mol (Ans 72.8 kJ/mol)

Step 1: By Hess's law, delta H = 2 x delta H1 + 2 x delta H2 - delta H3 (twice reaction i, twice reaction ii, minus reaction iii, so that CO2, H2O

and O2 cancel). Step 2: delta H = 2 x (-393.5) + 2 x (-285.7) - (-1430.9) = -787 - 571.4 + 1430.9 = 72.5 kJ/mol (the book prints 72.8 kJ/mol, the difference being rounding of the given heats). Answer: about 72.5-72.8 kJ/mol.

Project

Project. Determination of specific heat capacity of iron. Take two beakers and an iron bolt. Determine the weight of the bolt and record it. Place the bolt in a beaker containing water in it and leave for a while so as the temperature of the water and the bolt comes to an equilibrium and record it. Now, take another dry beaker and record its weight. Add 100 mL water in this beaker and weigh again. Find the weight of the water by difference. Heat the water up to 70 to 80 degrees C and transfer the water into a calorimeter. Record the temperature of the water T1 in calorimeter and transfer the bolt from cold water to the calorimeter and record the temperature of water in calorimeter carefully. The temperature will fall to a minimum value. Record it as T2. Calculate delta T = T1 - T2, heat delivered to the bolt by water qw = mw x sw x delta T where mw and sw are the mass and specific heat of the water, respectively. Note that the same amount of heat, (qb = qw), has been absorbed by the bolt. Thus, specific heat of the bolt, sb, can be calculated as sb = qb/mb.

Step 1: Weigh the iron bolt and place it in a beaker of water until bolt and water reach the same temperature; record this temperature. Step 2: Take a dry beaker, weigh it, add 100 mL of water, weigh again to find the mass of water, and heat the water to 70-80 degrees C. Step 3: Transfer the hot water to a calorimeter, record its temperature T1, then transfer the bolt from the cold water into the calorimeter and record the lowest temperature reached, T2. Step 4: Heat given out by water is qw = mw x

sw x (T1 - T2). Step 5: The same heat is absorbed by the bolt, so the specific heat of the bolt is sb = qb / (mb x delta T of the bolt). Step 6: Compare the result with the known specific heat of iron, about 0.45 J/g/K, and discuss heat losses.

Unit 12 — Electrochemistry

Exercise — Choose the Correct Option

Q1. Weak electrolyte in solution is

a) completely ionized

b) slightly ionized [Correct]

c) never ionized

d) destroyed

Q2. Which one of the following is a strong electrolyte in solution?

a) Ammonium hydroxide

b) Carbonic acid

c) Potassium iodide [Correct]

d) Acetic acid

Q3. In electrolytic cell the cathode has a charge

a) Positive

b) Negative [Correct]

c) Neutral

d) Zero

Q4. The oxidation number of Cl in $HClO_3$ is

a) -I

b) + 1

c) + 3

d) + 5 [Correct]

Q5. The oxidation number of magnesium in $MgCO_3$ is

a) +3

b) +2 [Correct]

c) +1

d) -I

Q6. Which one of the following is a reduction reaction

a) Br2 -> 2Br- [Correct]

b) Fe2+ -> Fe3+

c) Zn -> Zn2+

d) Sn2+ -> Sn4+

Q7. A cell in which a non-spontaneous redox reaction is carried out by passing an electric current is a/an

a) Galvanic cell

b) Voltaic cell

c) Daniell cell

d) Electrolytic cell [Correct]

Q8. Zinc rod acts as anode in the Daniell cell but acts as cathode when coupled with aluminum electrode, this is because the standard reduction potential of

a) Zn > Al [Correct]

b) Zn < Al

c) Zn = Al

d) Zn = 0

Q9. Electrolysis is a process in which the cations and anions liberated from electrolyte are

a) hydrated

b) hydrolyzed

c) charged

d) discharged [Correct]

Q10. A cell which produces electric current by a redox reaction is called a/an

a) Voltaic cell [Correct]

b) Electrolytic cell

c) Half-cell

d) Standard cell

Q11. The lead storage battery is a/an

a) Daniell cell

b) Voltaic cell [Correct]

c) Dry cell

d) Electrolytic cell

Q12. The electrode potential of the standard hydrogen electrode is chosen as

a) 0 V [Correct]

b) 1V

c) -1 V

d) 2V

Q13. A cathode has the reduction potential

a) Less than the anode

b) More than the anode [Correct]

c) same as that of anode.

d) zero

Exercise — Short Questions

Q1. What is the oxidation number of N in N_2O, NO_2, N_2O_3, HNO_3, and NH_4^+.

Step 1: In N_2O, $2x + (-2) = 0$, so $x = +1$. Step 2: In NO_2, $x + 2(-2) = 0$, so $x = +4$. Step 3: In N_2O_3, $2x + 3(-2) = 0$, so $x = +3$. Step 4: In HNO_3, +1

+ x + 3(-2) = 0, so x = +5. Step 5: In NH4+, x + 4(+1) = +1, so x = -3.

Answers: N2O +1, NO2 +4, N2O3 +3, HNO3 +5, NH4+ -3.

Q2. Na+ is an oxidizing but Na is a reducing agent, why?

Na+ is an oxidising agent because it can accept an electron and be reduced to Na, oxidising another substance in the process. Na metal is a reducing agent because it easily loses its single valence electron to become Na+, reducing another substance. The same element in different oxidation states therefore plays opposite roles.

Q3. The lead storage battery is rechargeable. Why?

In the lead storage battery the discharge reactions produce PbSO4 on both plates, and these reactions are reversible. On charging, an external current reverses them, converting PbSO4 back into Pb and PbO2, so the battery can be used again and again. A dry cell cannot be recharged because its reactions are not reversible in this way.

Q4. Why a solution of sugar is non-conductor but that of table salt is a good conductor of electricity?

Sugar dissolves as neutral molecules, which carry no charge, so its solution cannot conduct electricity. Table salt dissolves as Na+ and Cl- ions, which are free to move and carry current through the solution, so its solution is a good conductor.

Q5. NaCl is non-conductor in the solid state but is a good conductor in molten form. Why?

In solid NaCl the Na+ and Cl- ions are fixed in the crystal lattice and cannot move, so solid salt does not conduct. In molten NaCl the lattice breaks down and the ions become free to move, so the melt conducts electricity well.

Q6. During electrolysis of fused NaCl, sodium metal is collected at cathode and not at anode, why?

During electrolysis of fused NaCl, Na+ ions move towards the cathode (the negative electrode) and Cl- ions towards the anode. At the cathode Na+ gains an electron and becomes sodium metal, Na+ + e- → Na, so sodium is collected at the cathode, not at the anode, where chlorine gas is liberated instead.

Q7. SHE acts as anode with Cu electrode but acts as cathode when connected to Zn electrode.

The standard hydrogen electrode has a reduction potential of 0.00 V, copper has +0.34 V and zinc has -0.76 V. With copper, hydrogen has the lower potential, so it is oxidised and acts as the anode. With zinc, hydrogen has the higher potential, so it is reduced and acts as the cathode. Whether SHE is anode or cathode depends on the other electrode's potential.

Q8. In electroplating, the article to be plated is made cathode, why?

In electroplating the article is made the cathode (negative electrode) so that the metal ions in the solution move towards it and are reduced to metal atoms on its surface, forming a smooth coating. If the article were the anode, it would dissolve instead of gaining a coating.

Exercise — Numerical

Q1. Balance the following equation by Oxidation Number method: i) KMnO4 + H2S + H2SO4 -> KHSO4 + MnSO4 + S + H2O ii) Fe + V2O3 -> Fe2O3 + VO iii) MnO2 + 4HCl -> MnCl2 + Cl2 + H2O iv) Mg + HCl -> MgCl2 + H2 v) Cu + H2SO4 -> CuSO4 + SO2 + H2O vi) KMnO4 + KNO2 + H2SO4 -> MnSO4 + KNO3 + K2SO4 + H2O

Step 1: Find the change in oxidation number: Mn falls from +7 to +2 (gain of 5 electrons), S in H2S rises from -2 to 0 (loss of 2 electrons). Step 2: Balance electrons: 2 Mn atoms gain 10 electrons, 5 S atoms lose 10

electrons. Step 3: Balance the rest by inspection. Balanced equations: (i) 2KMnO4 + 5H2S + 4H2SO4 -> 2KHSO4 + 2MnSO4 + 5S + 8H2O. (ii) 2Fe + 3V2O3 -> Fe2O3 + 6VO. (iii) MnO2 + 4HCl -> MnCl2 + Cl2 + 2H2O. (iv) Mg + 2HCl -> MgCl2 + H2. (v) Cu + 2H2SO4 -> CuSO4 + SO2 + 2H2O. (vi) 2KMnO4 + 5KNO2 + 3H2SO4 -> 2MnSO4 + 5KNO3 + K2SO4 + 3H2O.

Q2. Balance the following redox equations by the half reaction method: i) Sn2+ + Fe3+ -> Sn4+ + Fe2+ (Acidic medium) ii) Zn + Cr2O7^2- + H+ -> Zn2+ + Cr3+ + H2O (Acidic medium) iii) H2O2 + MnO4- + H+ -> Mn2+ + O2 + H2O (Acidic medium) iv) Sn2+ + I2 -> Sn4+ + I- (Neutral)

Balanced equations by the half-reaction (ion-electron) method: (i) Sn2+ + 2Fe3+ -> Sn4+ + 2Fe2+ (Acidic medium). (ii) 3Zn + Cr2O7^2- + 14H+ -> 3Zn2+ + 2Cr3+ + 7H2O (Acidic medium). (iii) 5H2O2 + 2MnO4- + 6H+ -> 2Mn2+ + 5O2 + 8H2O (Acidic medium). (iv) Sn2+ + I2 -> Sn4+ + 2I- (Neutral). In each case the oxidation half-reaction is multiplied so that the electrons lost equal the electrons gained, then H+ and H2O are used to balance oxygen and hydrogen in acidic medium.

Q3. Given the following cells at 25 degrees C. (a) Write the cell reactions (b) Calculate the cell voltage: i) 1) Mg2+/Mg, E0Red = -2.38 V; 2) Cu2+/Cu, E0Red = +0.34 V. ii) 1) 2H+/H2, E0Red = 0.00 V; 2) Ag+/Ag, E0Red = +0.80 V

Step 1: (i) Mg/Cu cell: Mg is oxidised at the anode, Mg -> Mg2+ + 2e-, and Cu2+ is reduced at the cathode, Cu2+ + 2e- -> Cu. Cell reaction: Mg + Cu2+ -> Mg2+ + Cu. Cell voltage = E(cathode) - E(anode) = 0.34 - (-2.38) = 2.72 V. Step 2: (ii) H2/Ag cell: H2 is oxidised at the anode, H2 -> 2H+ + 2e-, and Ag+ is reduced at the cathode, Ag+ + e- -> Ag. Cell reaction: H2 + 2Ag+ -> 2H+ + 2Ag. Cell voltage = 0.80 - 0.00 = 0.80 V.

Exercise — Descriptive Questions

Q1(a). Define the term electrolysis. State and explain Faraday's laws of electrolysis.

Electrolysis is the decomposition of an electrolyte by passing an electric current through it. Faraday's first law states that the mass of a substance liberated at an electrode is directly proportional to the quantity of electricity passed, m = ZIt. Faraday's second law states that when the same quantity of electricity passes through different electrolytes, the masses liberated are proportional to their chemical equivalents (atomic mass divided by valency).

Q1(b). A certain amount of current is passed through CuSO4 solution for fifty minutes. The net amount of Cu deposited was found to be 25g. Calculate the current when Z = 1 and At. Mass of Cu is 63.5g.

Step 1: Moles of Cu deposited = 25 / 63.5 = 0.3937 mol. Step 2: Cu2+ needs 2 electrons per atom, so charge = 2 x 0.3937 x 96500 = 76000 C (approx). Step 3: Time = 50 minutes = 3000 s. Step 4: Current I = Q/t = 76000 / 3000 = 25.3 A. Answer: about 25 A (the question's printed phrase 'when Z = 1' is unclear in the scan; the Faraday calculation above uses the valency 2 of Cu2+).

Q2. What is an electrode potential? How it can be measured?

Electrode potential is the potential difference set up between a metal electrode and the solution of its ions, due to the tendency of the metal to lose electrons or of its ions to gain them. It is measured by connecting the electrode to a standard hydrogen electrode (SHE) through a salt bridge and a voltmeter, with all solutions at 1 mol/dm^3 and 25 degrees C; the SHE is assigned 0.00 V and the measured voltage is the electrode's potential.

Q3. What is the difference between the following? Give examples. i) Reversible and irreversible cells ii) Electrolytic and voltaic cell

Reversible cells can be reversed by a small opposing current, so their chemical reactions run backwards on charging; the lead storage battery is an example. Irreversible cells cannot be reversed, so once discharged they are discarded; the dry cell is an example. An electrolytic cell uses electrical energy to drive a non-spontaneous reaction, with the anode positive and the cathode negative. A voltaic (galvanic) cell produces electrical energy from a spontaneous reaction, with the anode negative and the cathode positive.

Q4. i) Explain the term oxidation and reduction with example. ii) What is oxidation state? Give the oxidation number of Mn in KMnO4, MnO2, and Mn2O7.

Oxidation is the loss of electrons (or increase in oxidation number), for example Zn -> Zn2+ + 2e-. Reduction is the gain of electrons (or decrease in oxidation number), for example Cu2+ + 2e- -> Cu. Oxidation state (oxidation number) is the apparent charge an atom would have if all its bonds were ionic. The oxidation number of Mn is: in KMnO4, +7; in MnO2, +4; in Mn2O7, +7.

Q5. What is an electrolytic cell? Explain Daniell cell in detail. a) Calculate the electrical energy obtained from a Daniell cell. b) What are the half-cell reaction? Give the half-cell reactions of the Nelson's cell for the production of NaOH.

An electrolytic cell is a cell in which electrical energy drives a non-spontaneous chemical reaction. The Daniell cell is a voltaic cell: a zinc rod in ZnSO4 solution and a copper rod in CuSO4 solution, joined by a salt bridge. At the anode Zn is oxidised, Zn -> Zn2+ + 2e-, and at the cathode Cu2+ is reduced, Cu2+ + 2e- -> Cu, giving the cell reaction Zn + Cu2+ -> Zn2+ + Cu. Its standard voltage is E0 = 0.34 - (-0.76) = 1.10 V, and the electrical energy available is W = nFE0 = 2 x 96500 x 1.10 =

212300 J per mole of reaction. In the Nelson cell for producing NaOH, brine is electrolysed: at the anode 2Cl- -> Cl2 + 2e-, and at the cathode 2H2O + 2e- -> H2 + 2OH-, so the products are chlorine, hydrogen and sodium hydroxide.

Q6. What are electrochemical cells? Explain it in detail.

Electrochemical cells convert chemical energy into electrical energy or vice versa. A voltaic cell has two half-cells joined by a salt bridge: oxidation at the anode releases electrons, which travel through an external wire to the cathode where reduction occurs, producing a current. An electrolytic cell has an external power supply that forces a non-spontaneous reaction, with oxidation at the positive anode and reduction at the negative cathode. Both kinds have two electrodes, an electrolyte, and a complete circuit.

Q7. Write down the types of batteries. Explain fuel cells?

Batteries are of three main types: primary batteries, which cannot be recharged (dry cell, alkaline cell); secondary batteries, which can be recharged (lead storage battery, nickel-cadmium battery); and fuel cells, which keep working as long as fuel is supplied. A fuel cell, such as the hydrogen-oxygen cell, combines H2 and O2 continuously: at the anode 2H2 + 4OH- -> 4H2O + 4e-, at the cathode O2 + 2H2O + 4e- -> 4OH-, giving water as the only product. Fuel cells are efficient, quiet and pollution-free, and are used in spacecraft and vehicles.

Project

Project. Take two lead pencils and sharp them from both the ends in order to expose the inner carbon cores. Take a beaker filled with 100 mL water acidified with sulphuric acid. Fix the two pencils in the beaker in vertical position about 2cm from each other. Connect the pencil cores with

electrical wires with the terminals of a 9 volt battery, one pencil with the positive pole and other with the negative pole of the battery. When the battery is switched on you will see bubbles coming from the lower ends of the pencil cores. What do you think about the nature of the bubbles? Which reactions are carried out at the pencil ends? Identify the anode and the cathode?

Step 1: Sharpen two lead pencils at both ends to expose the graphite cores, and fix them vertically about 2 cm apart in a beaker of water acidified with sulphuric acid. Step 2: Connect one pencil to the positive terminal of a 9 V battery and the other to the negative terminal, and switch on. Step 3: Bubbles appear at both lower ends: hydrogen at the negative pencil (cathode), $2H^+ + 2e^- \rightarrow H_2$, and oxygen at the positive pencil (anode), $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$. Step 4: The pencil joined to the positive pole is the anode and the one joined to the negative pole is the cathode; twice as much hydrogen as oxygen is collected, confirming the electrolysis of water.